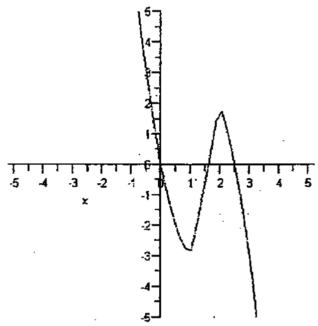
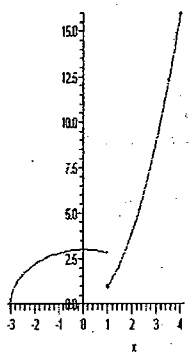
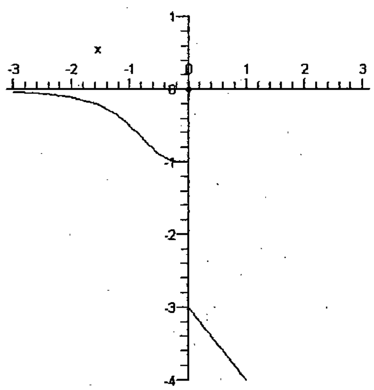
65. $\text{Dom}(s) = \mathbb{R}$, $\text{Rang}(s) = \mathbb{R}$.



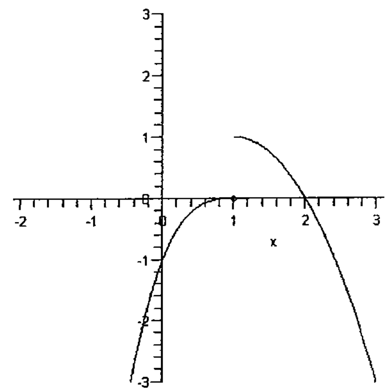
67. $\text{Dom}(g) = [-3,4]$, $\text{Rang}(g) = [0,16]$.



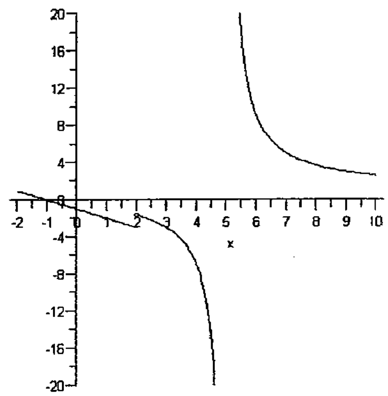
69. $\text{Dom}(i) = \mathbb{R} - \{0\}$, $\text{Rang}(i) = (-\infty,-3) \cup (-1,0)$.



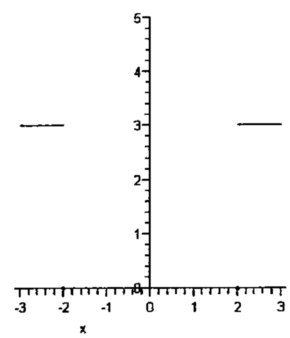
71. $\text{Dom}(k) = \mathbb{R}$, $\text{Rang}(k) = (-\infty,1)$



73. $\text{Dom}(m) = \mathbb{R} - \{5\}$, $\text{Rang}(m) = \mathbb{R}$

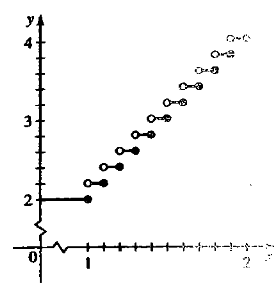


75. $\text{Dom}(p) = \mathbb{R}$, $\text{Rang}(p) = \{0,3\}$.



77.

$$C(x) = \begin{cases} 2 & \text{si} \quad 0 < x \leq 1 \\ 2.2 & \text{si} \quad 1 < x \leq 1.1 \\ 2.4 & \text{si} \quad 1.1 < x \leq 1.2 \\ \vdots & \qquad \vdots \\ 4.0 & \text{si} \quad 1.9 < x \leq 2 \end{cases}$$



79. (a) $E_y=[1,4)$ (b) $E_y=\left[\frac{1}{3},1\right)$ (c) $E_y=\left(\frac{\sqrt{11}}{2},\sqrt{3}\right]$

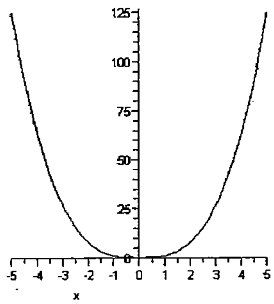
81. (a) 0,1 (b) $(0,1)$ (c) $(-\infty,0)\cup(1,+\infty)$

85. $f(x)=-\frac{7}{6}x-\frac{4}{3}$, $-2\le x\le 4$

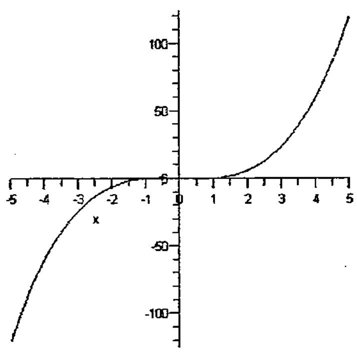
87. $f(x)=1+\sqrt{1-x}$, $x\le 1$

Sección 4.4

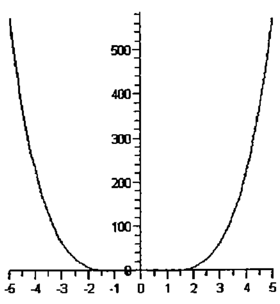
1. Par



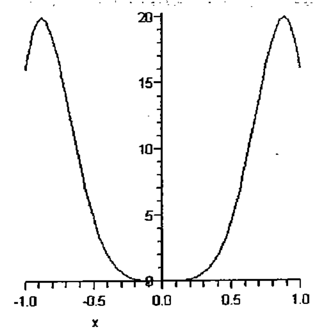
3. Impar.



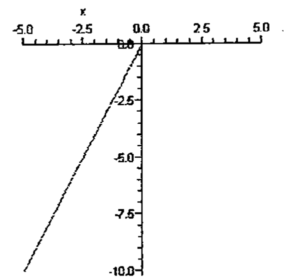
5. Par



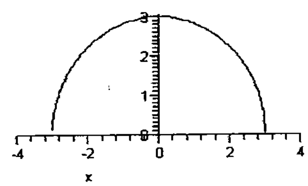
7. Par



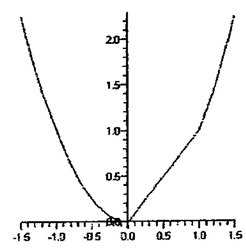
9. $\text{Dom}(f)=\mathbb{R}$, $\text{Rang}(f)=(-\infty,0]$, no es par, ni impar, ni periódica.



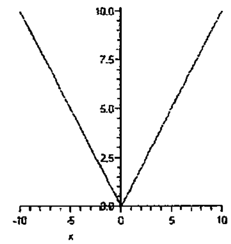
11. $\text{Dom}(f)=[-3,3]$, $\text{Rang}(f)=[0,3]$, es par.



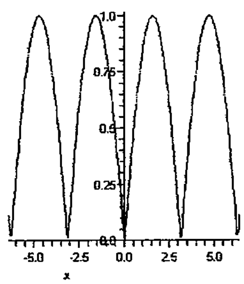
13. $\text{Dom}(f)=\mathbb{R}$, $\text{Rang}(f)=[0,+\infty)$, no es par, ni impar, ni peródica.



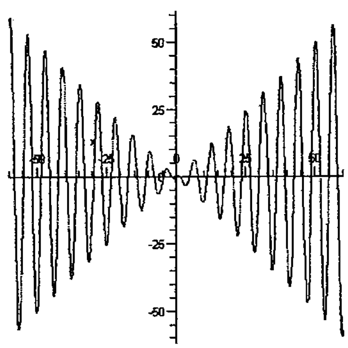
15. $\text{Dom}(f) = \mathbb{R}$, $\text{Rang}(f) = [0, +\infty)$, es par.



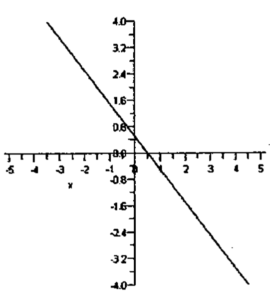
17. $\text{Dom}(f) = \mathbb{R}$, $\text{Rang}(f) = [0,1]$. Par y periódica de periodo π



19. $\text{Dom}(f) = \mathbb{R}$, $\text{Rang}(f) = \mathbb{R}$. No es par, ni impar, ni periódica.



21. $\text{Dom}(f) = \mathbb{R}$, $\text{Rang}(f) = \mathbb{R}$. No es par, ni impar, ni periódica.



23. Decreciente en $\left(-\infty, \frac{1}{4}\right]$, creciente en $\left[\frac{1}{4}, +\infty\right)$. En $x = \frac{1}{4}$ la función tiene un valor mínimo.

25. Decreciente en $(3, +\infty)$, creciente en $(-\infty, 3)$

27. Decreciente en $(4,6)$ $(6,7)$, creciente en $(3,4)$

29. Desplazamiento de la gráfica de f hacia la derecha 4 unidades.

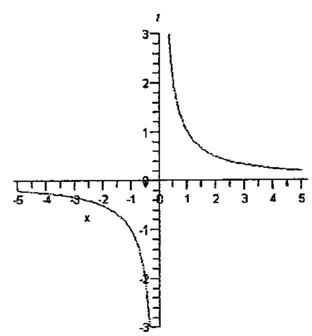
31. Desplazamiento de la gráfica de f hacia la izquierda 5 unidades.

33. Alargue de la gráfica de f vertical por un factor de 3.

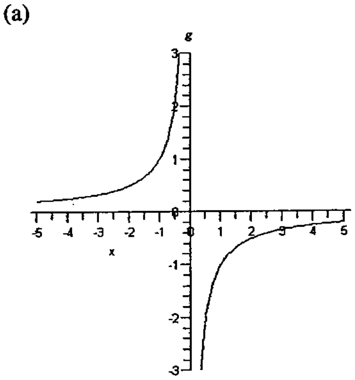
35. Reflexión de la gráfica de f en el eje Y.

37. Contracción horizontal de la gráfica de f por un factor de 1/2.

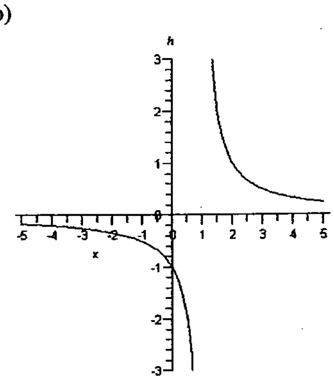
39.



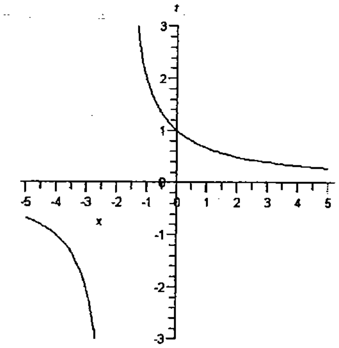
(a)



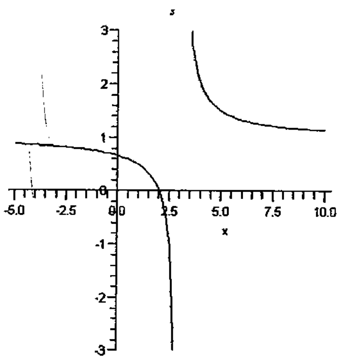
(b)



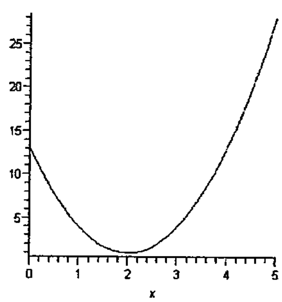
(c)



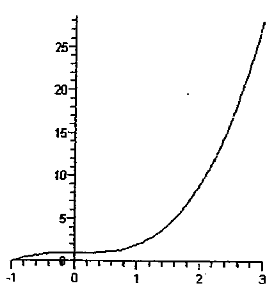
(d)



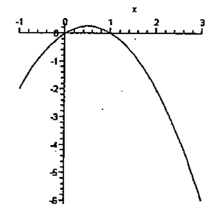
41. Decreciente en $(-\infty, 2)$, creciente en $(2, +\infty)$. Mínimo en $x = 2$.



43. Creciente en todo su dominio: $[-1,3]$. Mínimo en $x = -1$, máximo en $x = 3$.



45. Creciente en $\left(-1,\frac{1}{2}\right)$, decreciente en $\left(\frac{1}{2},3\right)$. Mínimo en $x = 3$ y máximo en $x = \frac{1}{2}$.



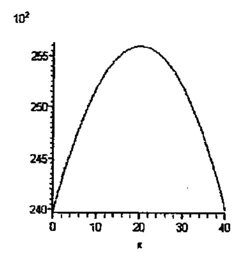
49. 30 veces.

51. $x = y = 50$

53. 1,200 m por 600 m.

55. 650 pesos

57. $p(x) = 24{,}000 + 160x - 4x^2$



80 árboles.

Sección 4.5

1. $(f+g)(x) = x^2 + 5$, $(f-g)(x) = x^2 - 2x - 5$, $(fg)(x) = x^3 + 4x^2 - 5x$, $\left(\dfrac{f}{g}\right)(x) = \dfrac{x^2 - x}{x+5}$. $\text{Dom}(f+g) = \text{Dom}(f-g) = \text{Dom}(fg) = \mathbb{R}$, $\text{Dom}(\frac{f}{g}) = \mathbb{R} - \{-5\}$

3. $(f+g)(x) = \sqrt{1+x} + \sqrt{1-x}$, $(f-g)(x) = \sqrt{1+x} - \sqrt{1-x}$, $(fg)(x) = \sqrt{1-x^2}$, $\left(\dfrac{f}{g}\right)(x) = \dfrac{\sqrt{1+x}}{\sqrt{1-x}}$. $\text{Dom}(f+g) = \text{Dom}(f-g) = \text{Dom}(fg) = [-1,1]$, $\text{Dom}(\frac{f}{g}) = [-1,1)$

5. $(f+g)(x) = \begin{cases} x+4 & \text{si } x<3 \\ \sqrt{x+3}+x+5 & \text{si } x>3 \end{cases}$,

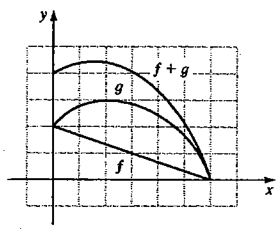
$(f-g)(x) = \begin{cases} -x-6 & \text{si } x<3 \\ \sqrt{x+3}-x-5 & \text{si } x>3 \end{cases}$,

$(fg)(x) = \begin{cases} -x-5 & \text{si } x<3 \\ \left(\sqrt{x+3}\right)(x-5) & \text{si } x>3 \end{cases}$,

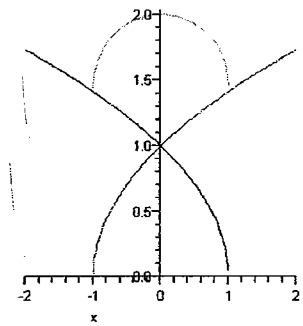
$\left(\dfrac{f}{g}\right)(x) = \begin{cases} \dfrac{-1}{x+5} & \text{si } x<3 \\ \dfrac{\sqrt{x+3}}{x+5} & \text{si } x>3 \end{cases}$.

$\mathrm{Dom}(f+g)=\mathrm{Dom}(f-g)=\mathrm{Dom}(fg)=\mathbb{R}-\{3\}$,
$\mathrm{Dom}(\frac{f}{g})=\mathbb{R}-\{-5,3\}$.

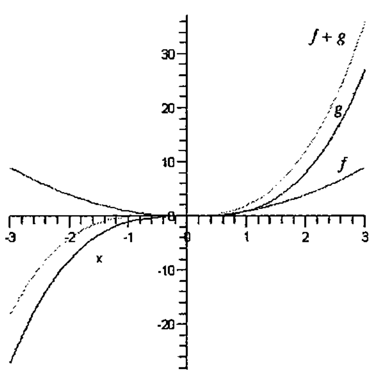
7.



9.



11.



13. a) $f(g(0))=-1$ y $g(f(0))=4$
b) $f(f(2))=119$ y $g(g(3))=0$
c) $(g\circ f)(-2)=340$ y $(f\circ g)(-2)=79$
d) $(g\circ f)(x)=64x^2-40x+4$, $(g\circ f)(x)=8x^2-24x-1$
e) $(g\circ g)(x)=x^4-6x^3+6x^2+9x$, $(f\circ f)(x)=64x-9$
f) $x^4-6x^3+70x^2-31x+4$

15. 5

17. 3

19. -2

13. $(f\circ g)(x)=8x+1$, $(g\circ f)(x)=8x+11$, $(g\circ g)(x)=16x-5$ y $(f\circ f)(x)=4x+9$
$\mathrm{Dom}(f\circ g)=\mathrm{Dom}(g\circ f)=\mathrm{Dom}(g\circ g)=\mathrm{Dom}(f\circ f)=\mathbb{R}$

23. $(f\circ g)(x)=\dfrac{1}{x^3+2x}$, $\mathrm{Dom}(f\circ g)=\mathbb{R}-\{-\sqrt{2},0,\sqrt{2}\}$,
$(g\circ f)(x)=\dfrac{1}{x^3}+\dfrac{2}{x}$, $\mathrm{Dom}(g\circ f)=\mathbb{R}-\{0\}$,
$(g\circ g)(x)=x^9+6x^7+12x^5+10x^3+4x$, $\mathrm{Dom}(g\circ g)=\mathbb{R}$ y
$(f\circ f)(x)=x$, $\mathrm{Dom}(f\circ f)=\mathbb{R}-\{0\}$.

25. $(f\circ g)(x)=\dfrac{1}{x-2}+2$, $\mathrm{Dom}(f\circ g)=\mathbb{R}-\{2\}$,
$(g\circ f)(x)=\dfrac{1}{x}$, $\mathrm{Dom}(g\circ f)=\quad-\{0\}$, $(g\circ g)(x)=\dfrac{x-2}{5-2x}$,
$\mathrm{Dom}(g\circ g)=\mathbb{R}-\left\{\dfrac{5}{2}\right\}$ y $(f\circ f)(x)=x+4$,$\mathrm{Dom}(f\circ f)=\mathbb{R}$.

27. $(f\circ g)(x)=\dfrac{2x+4}{x}+2$, $\mathrm{Dom}(f\circ g)=\mathbb{R}-\{0\}$,
$(g\circ f)(x)=\dfrac{1}{x+1}$, $\mathrm{Dom}(g\circ f)=\mathbb{R}-\{-1\}$,
$(g\circ g)(x)=\dfrac{x}{3x+4}$, $\mathrm{Dom}(g\circ g)=\mathbb{R}-\left\{-\dfrac{4}{3}\right\}$ y
$(f\circ f)(x)=x$, $\mathrm{Dom}(f\circ f)=\mathbb{R}-\{0\}$.

29. $(f\circ g)(x)=\sqrt{-x}$, $\mathrm{Dom}(f\circ g)=(-\infty,0]$, $(g\circ f)(x)=\sqrt{1-\sqrt{x^2-1}}$,
$\mathrm{Dom}(g\circ f)=\quad(g\circ g)(x)=\dfrac{x}{3x+4}$, $\mathrm{Dom}(g\circ g)=\mathbb{R}-\left\{-\dfrac{4}{3}\right\}$ y
$(f\circ f)(x)=x$, $\mathrm{Dom}(f\circ f)=\mathbb{R}-\{0\}$.

31. $(f\circ g)(x)=\dfrac{1}{\sqrt{x^2-4x}}$, $\mathrm{Dom}(f\circ g)=(-\infty,0)\cup(4,+\infty)$,
$(g\circ f)(x)=\dfrac{1}{x}-\dfrac{4}{\sqrt{x}}$, $\mathrm{Dom}(g\circ f)=\mathbb{R}_+$,
$(g\circ g)(x)=x^4-8x^3+12x^2+16x$, $\mathrm{Dom}(g\circ g)=\mathbb{R}$ y
$(f\circ f)(x)=\sqrt[4]{x}$, $\mathrm{Dom}(f\circ f)=\mathbb{R}_+\cup\{0\}$.

33. $(f\circ g)(x)=\begin{cases}\dfrac{1}{x+3} & \text{si } x<-6\\ (x+3)^2 & \text{si } x>-6\end{cases}$ $\mathrm{Dom}(f\circ g)=\mathbb{R}-\{-6\}$

$(g\circ f)(x)=\begin{cases}\dfrac{1}{x}+3 & \text{si } x<-3\\ x^2+3 & \text{si } x>-3\end{cases}$ $\mathrm{Dom}(g\circ f)=\mathbb{R}-\{-3\}$

$(g\circ g)(x)=x+6$, $\mathrm{Dom}(g\circ g)=\mathbb{R}$

$(f\circ f)(x)=\begin{cases}\dfrac{1}{x^2} & \text{si } x<-3\\ x^4 & \text{si } x>-3\end{cases}$ $\mathrm{Dom}(f\circ f)=\mathbb{R}-\{-3\}$

35. $(f\circ g)(x)=\begin{cases}-2 & \text{si} \quad x<0\\ 2 & \text{si} \quad x>0\end{cases}$ $\quad \text{Dom}(f\circ g)=\mathbb{R}-\{0\}$

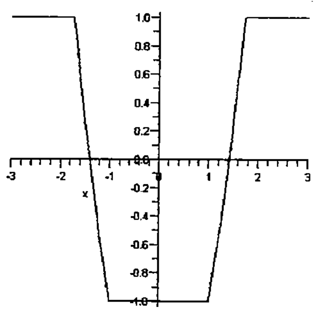
$(g\circ f)(x)=1 \quad \text{Dom}(g\circ f)=\mathbb{R}-\{0\}$

$(g\circ g)(x)=\begin{cases}-1 & \text{si} \quad x<0\\ 1 & \text{si} \quad x>0\end{cases}$ $\quad \text{Dom}(g\circ g)=\mathbb{R}-\{0\}$

$(f\circ f)(x)=\begin{cases}-\dfrac{x^4+3x^2+1}{(x^2+1)x} & \text{si} \quad x<0\\ \dfrac{x^4+3x^2+1}{(x^2+1)x} & \text{si} \quad x>0\end{cases}$ $\quad \text{Dom}(f\circ f)=\mathbb{R}-\{0\}$

37. $(a-1)d=b(c-1)$

39. $f(x)=\begin{cases}1 & \text{si} \quad x\le-\sqrt{3}\\ x^2-2 & \text{si} \quad -\sqrt{3}\le x\le-1\\ -1 & \text{si} \quad -1\le x\le 1\\ x^2-2 & \text{si} \quad 1\le x\le\sqrt{3}\\ 1 & \text{si} \quad x\ge\sqrt{3}\end{cases}$



41. $v(t)=\dfrac{4}{3}\pi t^2$

45. Si

47. No

49. Si

51. No

53. Si

55. (a) $[0,2)\cup(2,+\infty)\to(-\infty,0)\cup\left[\dfrac{1}{4},+\infty\right)$

(b) $(3,+\infty)\to(0,+\infty)$ (c) $\left[\dfrac{3}{2},+\infty\right)\to[0,+\infty)$

(d) $[0,+\infty)\to\left[\dfrac{3}{2},+\infty\right)$ (e) $[-4,+\infty)\to[0,+\infty)$

59. g es una restricción de f

61. Ninguna de las dos

63. $g:(-\infty,0]\to[0,+\infty);\ g^{-1}(x)=\sqrt{x}$

65. $f^{-1}(x)=\dfrac{1}{x}$

67. $f^{-1}(x)=6-x$

69. $f^{-1}(x)=\dfrac{3-x}{5}$

71. $f^{-1}(x)=\dfrac{2x+2}{1-x}$

73. $f^{-1}(x)=\sqrt[3]{\dfrac{5-x}{4}}$

75. $f^{-1}(x)=-\dfrac{1}{2}+\dfrac{1}{2}\sqrt{1+4x}$

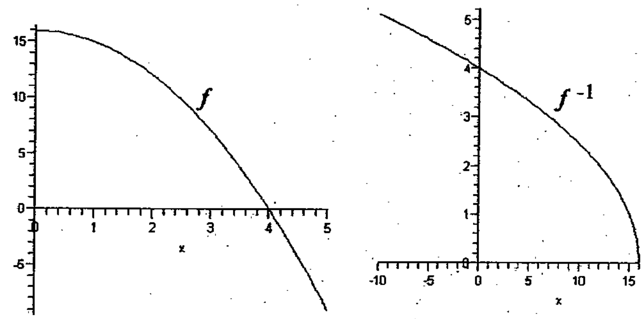
77. $f^{-1}(x)=\dfrac{x^2+1}{2}$

79. $f^{-1}(x)=\sqrt[3]{2-x^{1/5}}$

81. $f^{-1}(x)=\sqrt{9-x^2}$

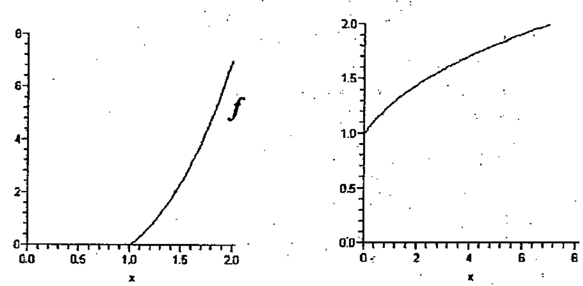
83. $f^{-1}(x)=-\sqrt{1-x}$

85.



$f^{-1}(x)=\sqrt{16-x}$

87.



$f^{-1}(x)=\sqrt[3]{x+1}$

89. $f(x)=\dfrac{x^2}{(1-x)^2}$

91. $f(x)=x^2-2$

Repaso

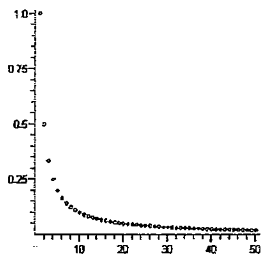
1. (a) $\frac{1}{25}$ unidades de temperatura (b) $f(x)=t(x,0)=\frac{1}{|x|}$,

$\mathrm{Dom}(f)=(0,50]$ (c) $g(y)=t(0,y)=\frac{1}{|y|}$, $\mathrm{Dom}(g)=(0,30]$

(d) $h(x)=t(x,x)=\frac{\sqrt{2}}{2|x|}$, $\mathrm{Dom}(h)=(0,30]$.

3. (a) $\mathrm{Dom}(f)=\mathbb{N}$ (b) $Rang(f)=\left\{\frac{1}{n}\mid n\in\mathbb{N}\right\}\subseteq(0,1]$

(c) $G_f=\left\{(n,\frac{1}{n})\mid n\in\mathbb{N}\right\}$



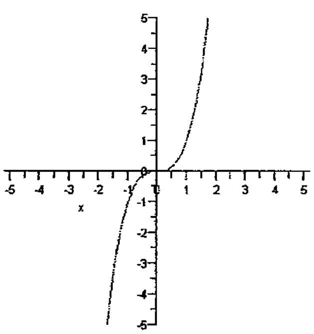
(d) $\lim_{n\to\infty}\frac{1}{n}=0$

Examen

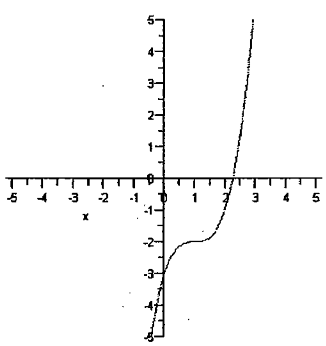
(a), (b) son gráficas de funciones (a) es inyectiva

1. $[0,1)\cup(1,+\infty)$

2. (a)



(b)



3. (a) Desplazar 3 unidades hacia la izquierda, reflejar en el eje y y luego desplazar 2 unidades hacia arriba.
(b) Reflejar en el eje x y después en el eje y.

4. (a) $f(-2)=-3, f\left(\frac{1}{2}\right)=-1, f(3)=7$ (b) No

5. (a) gráfica a) f, gráfica b) j, gráfica c) k

(b) No es inyectiva, $\begin{array}{lccc} g: & (-\infty,0] & \to & (-\infty,1] \\ & x & \mapsto & g(x)=x^2+1 \end{array}$,

$\begin{array}{lccc} g^{-1}: & (-\infty,1] & \to & (-\infty,0] \\ & x & \mapsto & g^{-1}(x)=-\sqrt{1-x} \end{array}$

(c) $-\sqrt{2},\sqrt{2}$

6. (a) $(f+g)(x)=\begin{cases} 4-x^2+2x & \text{si } x<-\frac{1}{2} \\ 2-x^2-2x & \text{si } x\geq-\frac{1}{2} \end{cases}$, $\mathrm{Dom}(f+g)=\mathbb{R}$

(b) $(fg)(x)=\begin{cases} 4+4x-2x^2-2x^3 & \text{si } x<-\frac{1}{2} \\ -4x+2x^3 & \text{si } x\geq-\frac{1}{2} \end{cases}$, $\mathrm{Dom}(fg)=\mathbb{R}$

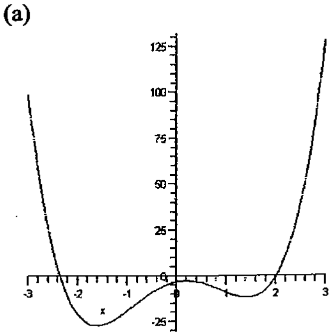
(c) $\left(\frac{f}{g}\right)(x)=\begin{cases} \frac{2+2x}{x^2-2} & \text{si } x<-\frac{1}{2} \\ \frac{2x}{x^2-2} & \text{si } x\geq-\frac{1}{2} \end{cases}$, $\mathrm{Dom}(fg)=\mathbb{R}-\left\{-\sqrt{2},\sqrt{2}\right\}$

(d) $(f\circ g)(x)=\begin{cases} 6-2x^2 & \text{si } x<-\frac{\sqrt{10}}{2} \\ 2x^2-4 & \text{si } -\frac{\sqrt{10}}{2}\leq x\leq\frac{\sqrt{10}}{2} \\ 6-2x^2 & \text{si } x>\frac{\sqrt{10}}{2} \end{cases}$

(e) $(g\circ f)(x)=\begin{cases} -2-8x-4x^2 & \text{si } x<-\frac{1}{2} \\ 2-4x^2 & \text{si } x\geq-\frac{1}{2} \end{cases}$

1. (a) $A(x)=400x-2x^2$ (b) 200 pies por 100 pies

2. (a)



(b) No (c) Mínimo local ≈ -27.18 cuando $x \approx -1.61$; Maximo local ≈ -2.55 cuando $x \approx 0.18$; Mínimo local ≈ -11.93 cuando $x \approx -1.43$

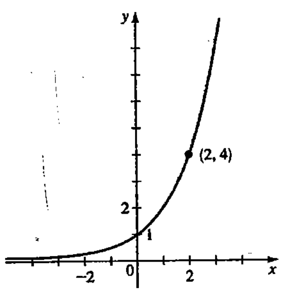
(d) $[-27.18, +\infty)$

(c) Creciente en $[-1.61, 0.18] \cup [1.43, +\infty)$, decreciente en $(-\infty, -1.61] \cup [0.18, 1.43]$

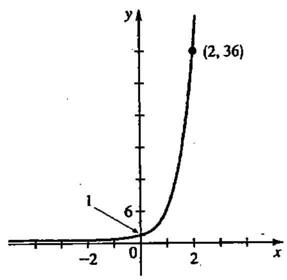
CAPÍTULO 5

Sección 5.1

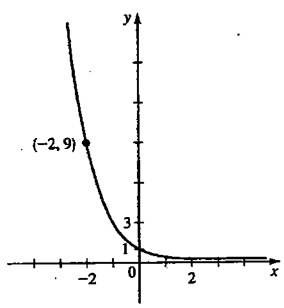
1.



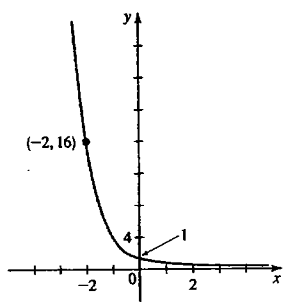
3.



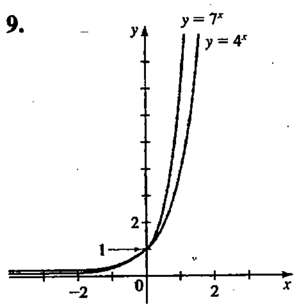
5.



7.



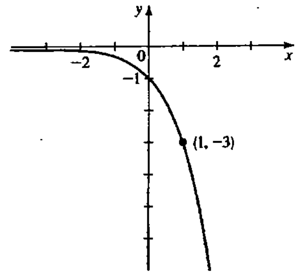
9.



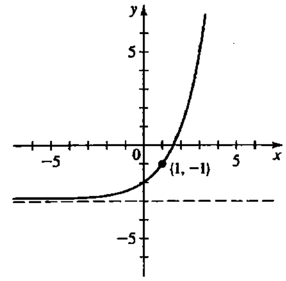
11. $f(x) = 3^x$ **13.** $f(x) = \left(\frac{1}{4}\right)^x$

15. III **17.** I **19.** II

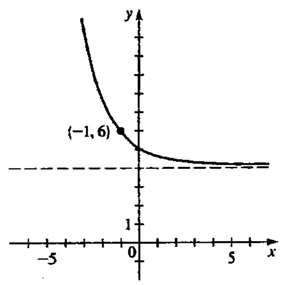
21. $\mathbb{R}$, $(-\infty, 0)$, $y = 0$



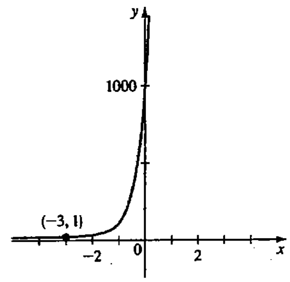
23. $\mathbb{R}$, $(-3, \infty)$, $y = -3$



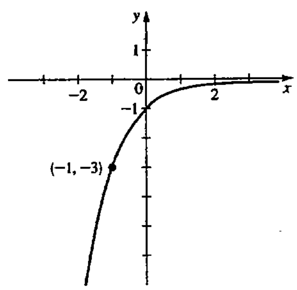
25. $\mathbb{R}$, $(4, \infty)$, $y = 4$



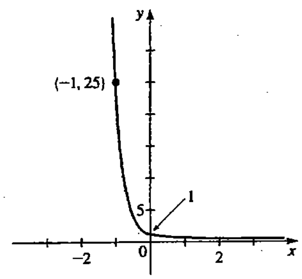
27. $\mathbb{R}$, $(0, \infty)$, $y = 0$



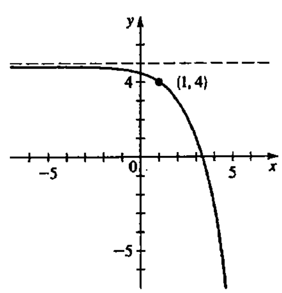
29. $\mathbb{R}$, $(-\infty, 0)$, $y = 0$



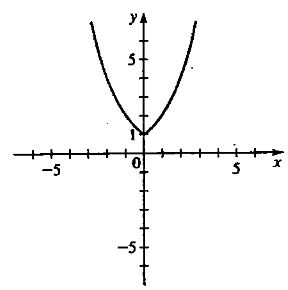
31. $\mathbb{R}$, $(0, \infty)$, $y = 0$



33. $\mathbb{R}$, $(-\infty, 5)$, $y = 5$

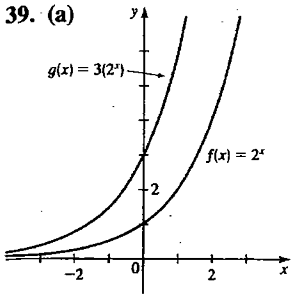


35. $\mathbb{R}$, $[1, \infty)$, no tiene asíntota



37. $y = 3(2^x)$

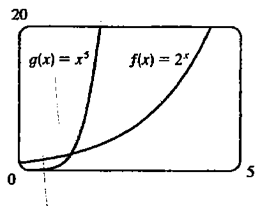
39. (a)



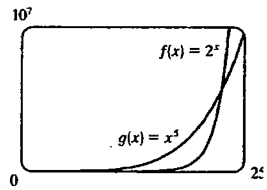
(b) La gráfica de g tiene mayor pendiente que la de f.

43. (a)

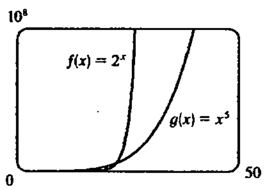
(i)



(ii)



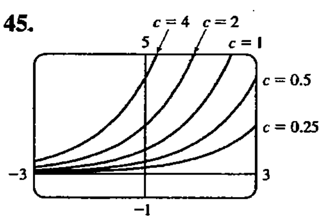
(iii)



La gráfica de f aumenta al final con mucha mayor rapidez que la de g.

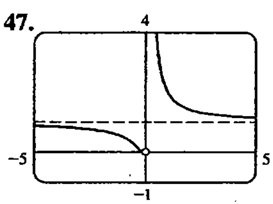
(b) 1.2, 22.4

45.



Cuanto mayor es el valor de c, la gráfica aumenta con mayor rapidez.

47.



Asíntota vertical: $x = 0$, asíntota horizontal: $y = 1$

49. Mínimo local en $\approx (0.37, 0.69)$

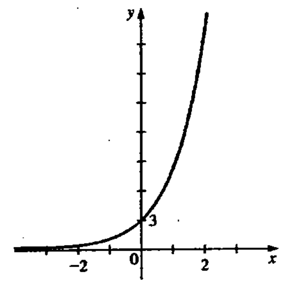
51. (a) Creciente en $(-\infty, 0.50]$, decreciente en $[0.50, \infty)$

(b) $(0, 1.78]$

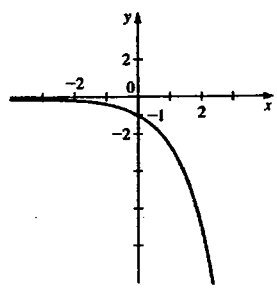
Sección 5.2

1.

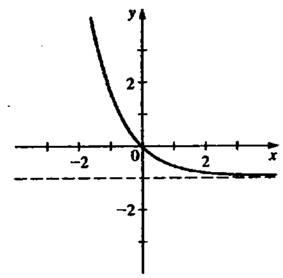
x	$f(x) = 3e^x$
−2	0.4
−1.5	0.7
−1	1.1
−0.5	1.8
0	3
0.5	4.9
1	8.2
1.5	13.4
2	22.2



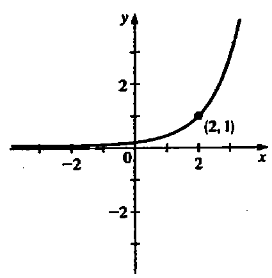
3. $\mathbb{R}$, $(-\infty, 0)$, $y = 0$



5. $\mathbb{R}$, $(-1, \infty)$, $y = -1$



7. $\mathbb{R}$, $(0, \infty)$, $y = 0$



9. (a) \$16,288.95 **(b)** \$26,532.98 **(c)** \$43,219.42

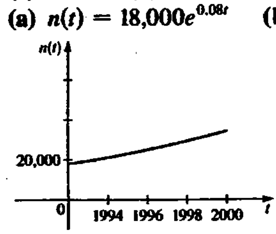
11. (a) \$4,615.87 **(b)** \$4,658.91 **(c)** \$4,697.04 **(d)** \$4,703.11 **(e)** \$4,704.68 **(f)** \$4,704.93 **(g)** \$4,704.94

13. (i) **15.** \$7,678.96

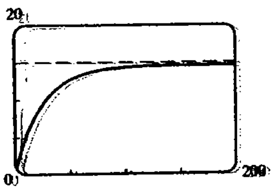
17. (a) 45% **(b)** 500 **(c)** 4744

19. (a) $n(t) = 18{,}000e^{0.08t}$ **(b)** 34,137

(c)

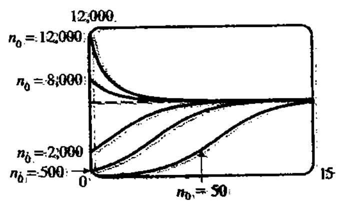


21. **(a)** 233 millones **(b)** 181 millones
23. 5,870 billones **25.** **(a)** Unos 500 **(b)** 361,000
27. **(a)** 6 g **(b)** 1 g
29. **(a)** 2.7 lb **(b)** 4.9 lb
(c)

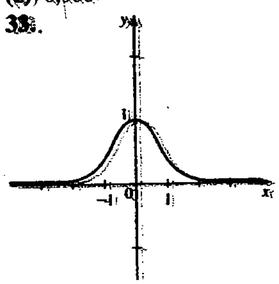


(d) 15 lb

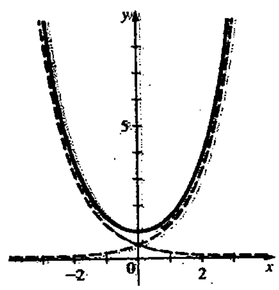
31. **(a)** 5164 **(b)**



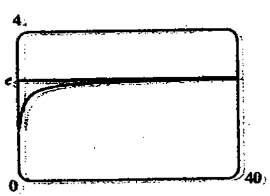
(c) 6,000
33.



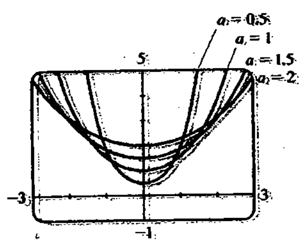
35.



41.



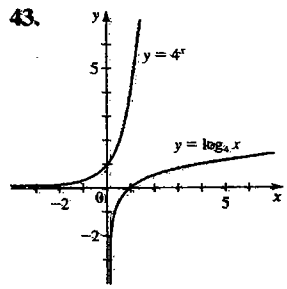
43. **(a)**



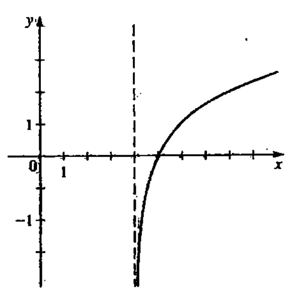
(b) Mientras mayor es el valor de a, la gráfica tiene menos pendiente y su valor mínimo es menor
45. Máximo local en $(1.00, 0.37)$

Sección 5.3

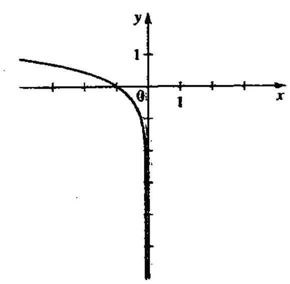
1. **(a)** $2^5 = 32$ **(b)** $5^0 = 1$
3. **(a)** $4^{1/2} = 2$ **(b)** $2^{-4} = \frac{1}{16}$
5. **(a)** $e^x = 5$ **(b)** $e^5 = y$
7. **(a)** $\log_2 8 = 3$ **(b)** $\log_{10} 0.001 = -3$
9. **(a)** $\log_4 0.125 = -\frac{3}{2}$ **(b)** $\log_7 343 = 3$
11. **(a)** $\ln 2 = x$ **(b)** $\ln y = 3$
13. **(a)** 4 **(b)** 3 **(c)** 1 **15.** **(a)** 2 **(b)** 2 **(c)** 10
17. **(a)** -3 **(b)** $\frac{1}{2}$ **(c)** -1
19. **(a)** 37 **(b)** 8 **(c)** $\sqrt{5}$
21. **(a)** $-\frac{2}{3}$ **(b)** 4 **(c)** -1 **23.** **(a)** 32 **(b)** 4
25. **(a)** 100 **(b)** 25 **27.** **(a)** 2 **(b)** 4
29. **(a)** 0.3010 **(b)** 1.5465 **(c)** -0.1761
31. **(a)** 1.6094 **(b)** 3.2308 **(c)** 1.0051
33. $y = \log_5 x$ **35.** $y = \log_9 x$
37. II **39.** III **41.** VI
43.



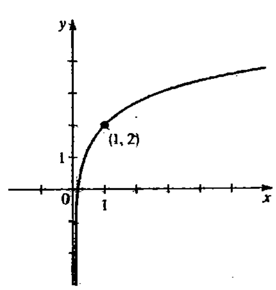
45. $(4, \infty)$, $\mathbb{R}$, $x = 4$



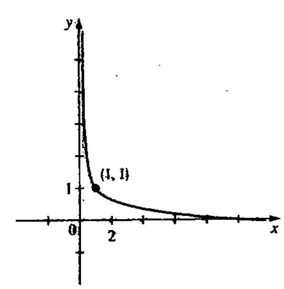
47. $(-\infty, 0)$, $\mathbb{R}$, $x = 0$



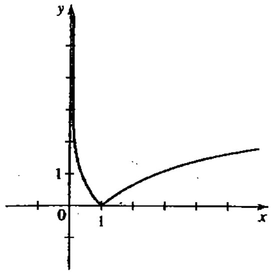
49. $(0, \infty)$, $\mathbb{R}$, $x = 0$



51. $(0, \infty)$, $\mathbb{R}$, $x = 0$

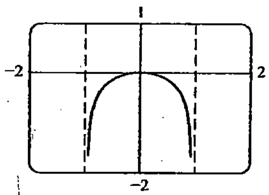


53. $(0,\infty)$, $[0,\infty)$, $x = 0$



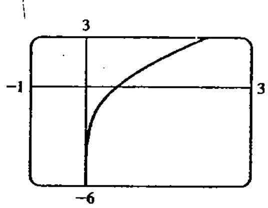
55. $\left(-\frac{2}{3},\infty\right)$ **57.** $(-\infty,-1)\cup(1,\infty)$ **59.** $(0,2)$

61.



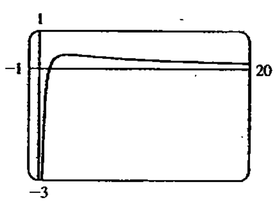
dominio = (−1, 1)
asíntotas verticales:
$x = 1, x = -1$
máximo local en (0, 0)

63.



dominio = $(0,\infty)$
asíntota vertical: $x = 0$
no tiene máximo ni mínimo

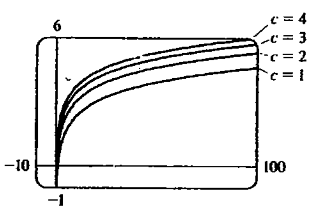
65.



dominio = $(0,\infty)$
asíntota vertical: $x = 0$
asíntota horizontal: $y = 0$
máximo local en (2.72, 0.37)

67. La gráfica de f aumenta con menos rapidez que la de g.

69. **(a)**



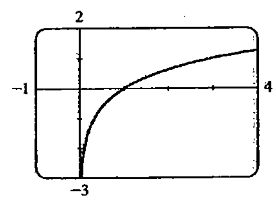
(b) La gráfica de $f(x) = \log(cx)$ es la de $f(x) = \log x$ desplazada $\log c$ unidades hacia arriba.

71. **(a)** $(1,\infty)$ **(b)** $f^{-1}(x) = 10^{2^x}$

73. **(a)** $f^{-1}(x) = \log_2\left(\dfrac{x}{1-x}\right)$ **(b)** $(0,1)$

Sección 5.4

1. $\log_2 x + \log_2(x-1)$ **3.** $23\log 7$
5. $\log_2 A + 2\log_2 B$ **7.** $\log_3 x + \frac{1}{2}\log_3 y$
9. $\frac{1}{3}\log_5(x^2+1)$ **11.** $\frac{1}{2}(\ln a + \ln b)$
13. $3\log x + 4\log y - 6\log z$
15. $\log_2 x + \log_2(x^2+1) - \frac{1}{2}\log_2(x^2-1)$
17. $\ln x + \frac{1}{2}(\ln y - \ln z)$ **19.** $\frac{1}{4}\log(x^2+y^2)$
21. $\frac{1}{2}[\log(x^2+4) - \log(x^2+1) - 2\log(x^3-7)]$
23. $\frac{1}{2}\ln x + 4\ln z - \frac{1}{3}\ln(y^2+6y+17)$
25. $\frac{3}{2}$ **27.** 1 **29.** 3 **31.** $\ln 8$ **33.** 16
35. $\log_3 160$ **37.** $\log_2(AB/C^2)$ **39.** $\log\left[\dfrac{x^4(x-1)^2}{\sqrt[3]{x^2+1}}\right]$
41. $\ln[5x^2(x^2+5)^3]$
43. $\log\left[\sqrt[3]{2x+1}\sqrt{(x-4)/(x^4-x^2-1)}\right]$
45. 2.807355 **47.** 2.182658 **49.** 0.655407
51. 4.165458
53.



Sección 5.5

1. 1.7227 **3.** −0.5850 **5.** 1.2040 **7.** 0.0767
9. 0.2524 **11.** 1.9349 **13.** −43.0677
15. 2.1492 **17.** 6.2126 **19.** −2.9469
21. −2.4423 **23.** 14.0055 **25.** ± 1 **27.** $0, \frac{4}{3}$
29. $\ln 2 \approx 0.6931$, 0 **31.** $\frac{1}{2}\ln 3 \approx 0.5493$
33. $e^{10} \approx 22026$ **35.** 0.01 **37.** $\frac{95}{3}$
39. $3 - e^2 \approx -4.3891$ **41.** 5 **43.** 5 **45.** $\frac{13}{12}$
47. 6 **49.** $\frac{3}{2}$ **51.** $1/\sqrt{5} \approx 0.4472$ **53.** 13 días
55. **(a)** $t = -\frac{5}{13}\ln\left(1 - \frac{13}{60}I\right)$ **(b)** 0.218 s
57. 2.21 **59.** 0.00, 1.14 **61.** −0.57 **63.** 0.36
65. $2 < x < 4$ o $7 < x < 9$ **67.** $\log 2 < x < \log 5$
69. 101, 1.1 **71.** $\log_2 3 \approx 1.58$

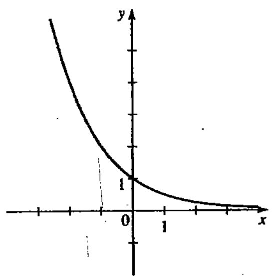
Sección 5.6

1. **(a)** \$12,870.19 **(b)** 8.24 años **3.** 5 años
5. 8.15 años **7.** 8.30%
9. **(a)** 500 **(b)** 45% **(c)** 1,929 **(d)** 6.66 h
11. **(a)** $n(t) = 112{,}000e^{0.04t}$ **(b)** Más o menos 142,000 **(c)** 2,008
13. **(a)** 20,000 **(b)** $n(t) = 20{,}000e^{0.1096t}$
(c) Más o menos 48,000 **(d)** 2004
15. **(a)** $n(t) = 8600e^{0.1508t}$ **(b)** Más o menos 11,600 **(c)** 4.6 h
17. **(a)** 2029 **(b)** 2049 **19.** 22.85 h
21. **(a)** $n(t) = 10e^{-0.0231t}$ **(b)** 1.6 g **(c)** 70 años

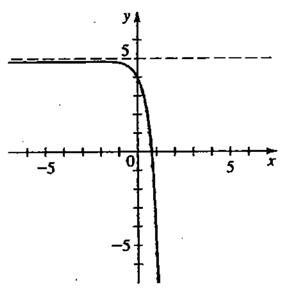
23. 16 años **25.** 149 h **27.** 3560 años
29. **(a)** 210 °F **(b)** 153 °F **(c)** 28 min
31. **(a)** 137 °F **(b)** 116 min
33. **(a)** 2.3 **(b)** 3.5 **(c)** 8.3
35. **(a)** 10^{-3} M **(b)** 3.2×10^{-7} M **37.** $4.8 \leq \text{pH} \leq 6.4$
39. $\log 20 \approx 1.3$ **41.** Doble intensidad **43.** 8.2
45. 6.3×10^{-3} W/m^2 **47.** **(b)** 106 dB

Repaso del capítulo 5

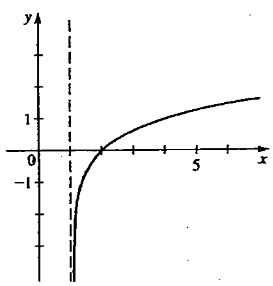
1. $\mathbb{R}, (0, \infty), y = 0$



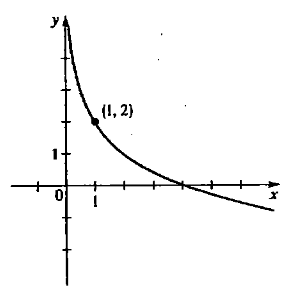
3. $\mathbb{R}, (-\infty, 5), y = 5$



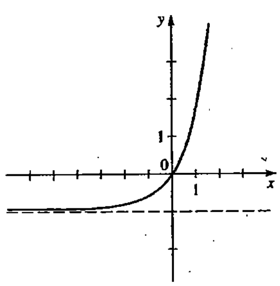
5. $(1, \infty), \mathbb{R}, x = 1$



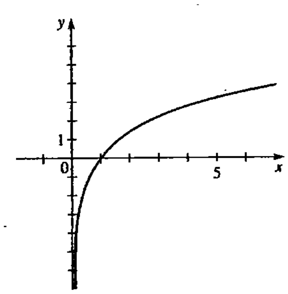
7. $(0, \infty), \mathbb{R}, x = 0$



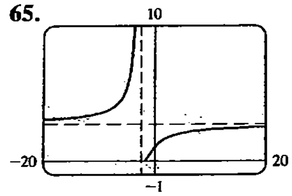
9. $\mathbb{R}, (-1, \infty), y = -1$



11. $(0, \infty), \mathbb{R}, x = 0$

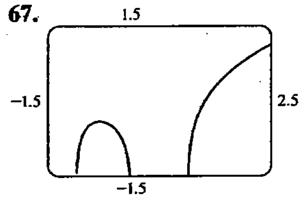


13. $\left(-\infty, \frac{1}{2}\right)$ **15.** $2^{10} = 1024$ **17.** $10^y = x$
19. $\log_2 64 = 6$ **21.** $\log 74 = x$ **23.** 7 **25.** 45
27. 6 **29.** -3 **31.** $\frac{1}{2}$ **33.** 2 **35.** 92 **37.** $\frac{2}{3}$
39. $\log A + 2 \log B + 3 \log C$
41. $\frac{1}{2}[\ln(x^2 - 1) - \ln(x^2 + 1)]$
43. $2 \log_5 x + \frac{3}{2} \log_5(1 - 5x) - \frac{1}{2} \log_5(x^3 - x)$
45. $\log 96$ **47.** $\log_2\left[\dfrac{(x - y)^{3/2}}{(x^2 + y^2)^2}\right]$ **49.** $\log\left(\dfrac{x^2 - 4}{\sqrt{x^2 + 4}}\right)$
51. -15 **53.** $\frac{1}{3}(5 - \log_5 26) \approx 0.99$
55. $\frac{4}{3} \ln 10 \approx 3.07$ **57.** 3 **59.** $-4, 2$ **61.** 0.430618
63. 2.303600
65.



asíntota vertical
$x = -2$
asíntota horizontal
$y = 2.72$
sin máximo ni mínimo

67.

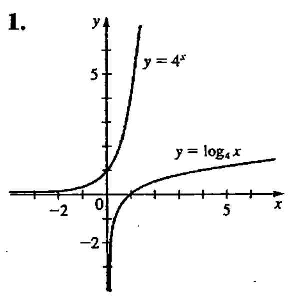


asíntotas verticales:
$x = -1, x = 0, x = 1$
máximo local en
$(-0.58, -0.41)$

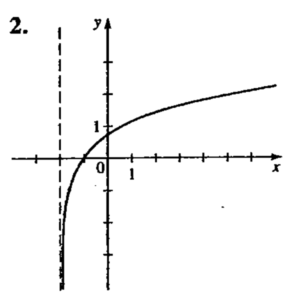
69. 2.42 **71.** $0.16 < x < 3.15$
73. Creciente en $(-\infty, 0]$ y en $[1.10, \infty)$, decreciente en $[0, 1.10]$
75. 1.953445 **77.** $\log_4 258$
79. **(a)** \$16,081.15 **(b)** \$16,178.18 **(c)** \$16,197.64 **(d)** \$16,198.31
81. **(a)** $n(t) = 30e^{0.15t}$ **(b)** 55 **(c)** 19 años
83. **(a)** 9.97 mg **(b)** 1.39×10^5 años
85. **(a)** $n(t) = 150e^{-0.0004359t}$ **(b)** 97.0 mg **(c)** 2,520 años
87. **(a)** $n(t) = 1500e^{0.1515t}$ **(b)** 7940 **89.** 7.9, básica
91. 8.0

Examen del capítulo 5

1.



2.



$(-2, \infty), \mathbb{R}, x = -2$

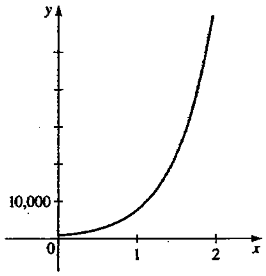
3. **(a)** $\frac{3}{2}$ **(b)** 3 **(c)** $\frac{2}{3}$ **(d)** 2
4. $\frac{1}{2}[\log(x^2 - 1) - 3 \log x - 5 \log(y^2 + 1)]$

5. $\ln\left[\dfrac{x\sqrt{3-x^4}}{(x^2+1)^2}\right]$

6. **(a)** 4.32 **(b)** 0.77 **(c)** 5.39 **(d)** 2

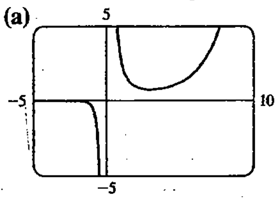
7. **(a)** $n(t) = 1{,}000e^{2.07944t}$ **(b)** 22,627 **(c)** 1.3 h

(d)



8. 8.33 años **9.** $\left(-4, \frac{8}{5}\right)$

10. **(a)**



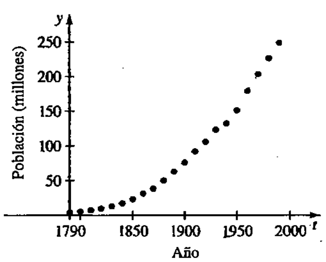
(b) $x = 0, y = 0$

(c) Mínimo local en $(3.00, 0.74)$

(d) $(-\infty, 0) \cup [0.74, \infty)$ **(e)** $-0.85, 0.96, 9.92$

ENFOQUE EN MODELADO

1. **(a)**

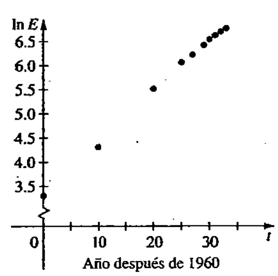


(b) $y = ab^t$, siendo $a = 4.041807 \times 10^{-16}$ y $b = 1.021003194$, y y es la población en el año t, en millones.

(c) 457.9 millones **(d)** 221.2 millones **(e)** No

3. **(a)** Sí

(b) Sí, la gráfica de dispersión parece ser lineal



(c) $\ln E = 3.30161 + 0.10769t$, siendo t años a partir de 1960 y E el gasto en miles de millones de dólares.

(d) $E = 27.15633e^{0.10769t}$

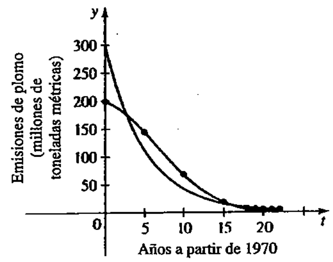
(e) 1,310,900 billones de dólares

5. **(a)** $y = ab^t$, siendo $a = 301.813054$, $b = 0.819745$ y t la cantidad de años a partir de 1970

(b) $y = at^4 + bt^3 + ct^2 + dt + e$, siendo $a = -0.002430$, $b = 0.135159$, $c = -2.014322$, $d = -4.055294$, $e = 199.092227$, y t la cantidad de años a partir de 1970

(c) En las gráficas se ve que el modelo de polinomio de cuarto grado es mejor.

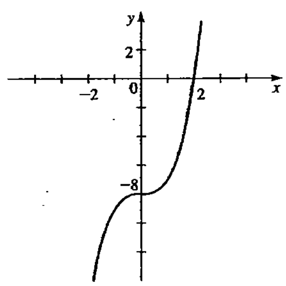
(d) 202.8, 27.8; 184.0, 43.5



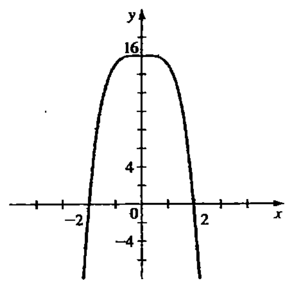
CAPÍTULO 6

Sección 6.1

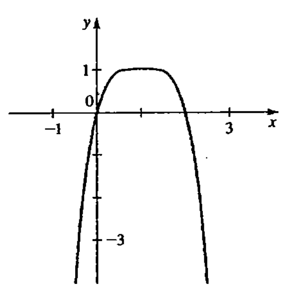
1.



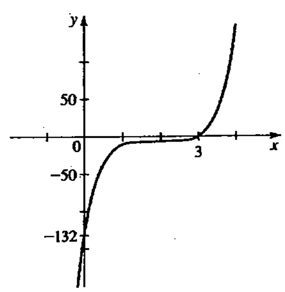
3.



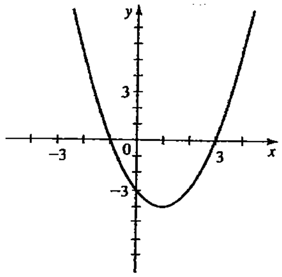
5.



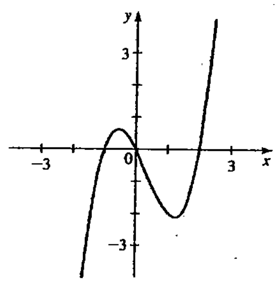
7.



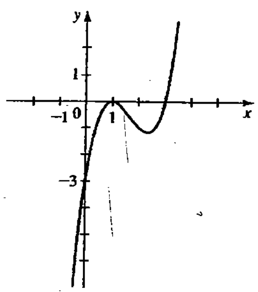
9.



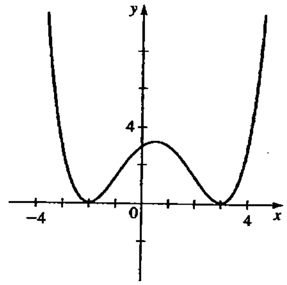
11.



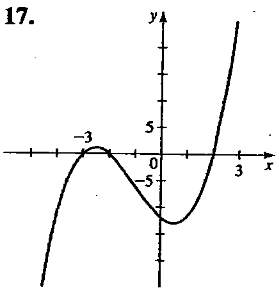
13.



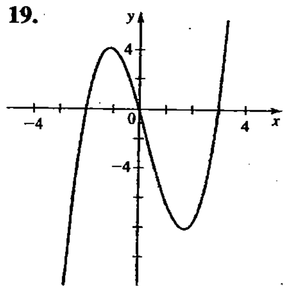
15.



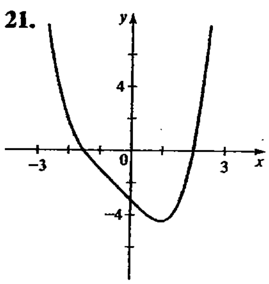
17.



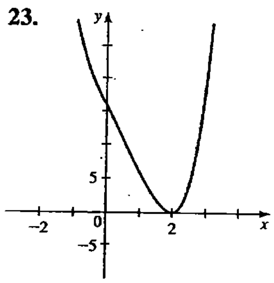
19.



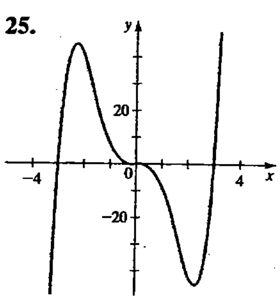
21.



23.

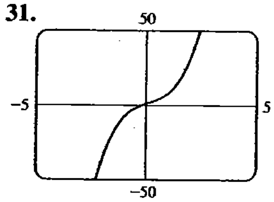


25.



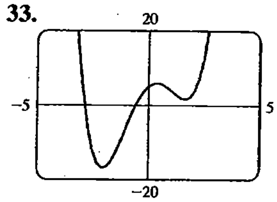
27. III **29.** IV

31.



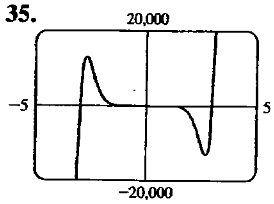
$y \to \infty$ cuando $x \to \infty$
$y \to -\infty$ cuando $x \to -\infty$

33.



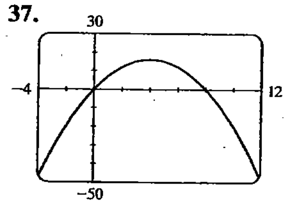
$y \to \infty$ cuando $x \to \infty$

35.



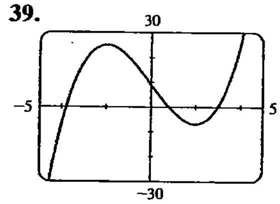
$y \to \infty$ cuando $x \to \infty$
$y \to -\infty$ cuando $x \to -\infty$

37.



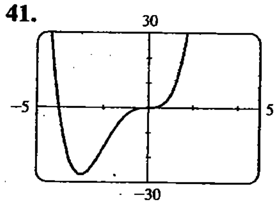
intersecciones en x 0, 8
intersección en y 0
máximo local en (4, 16)

39.



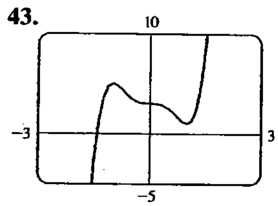
intersecciones en x −3.79, 0.79, 3
intersección en y 9
máximo local en (−2, 25)
mínimo local en (2, −7)

41.



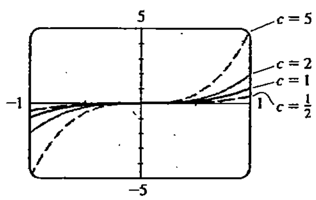
intersecciones en x −4, 0
intersección en y 0
mínimo local en (−3, −27)

43.



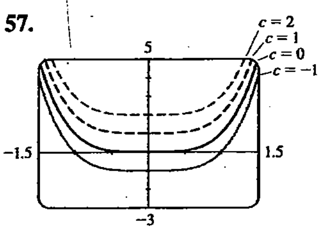
intersecciones en x −1.42
intersección en y 3
máximo local en (−1, 5)
mínimo local en (1, 1)

45. Máximo local en $(0.75, 6.13)$□
47. Máximo local en $(-0.33, 0.19)$,□
mínimo local en $(1.00, -1.00)$□
49. Máximo local en $(0, 4)$;□
mínimos locales en $(-1.58, -2.25)$, $(1.58, -2.25)$□
51. No hay extremos□
53. Máximo local: $(0.44, 0.33)$;□
mínimos locales en $(1.09, -1.15)$, $(-1.12, -3.36)$
55.



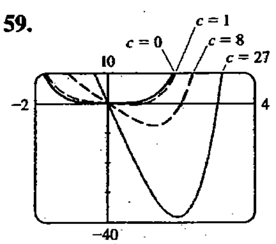
Al aumentar el valor de c la gráfica se alarga verticalmente.

57.



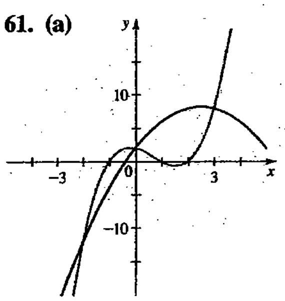
Al aumentar el valor de c la gráfica se mueve hacia arriba.

59.



Al aumentar el valor de c hay una bajada más pronunciada en el 4o. cuadrante y la intersección en x positiva se recorre hacia la derecha.

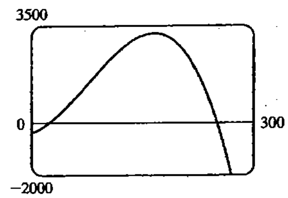
61. (a)



(b) Tres
(c) $(0, 2)$, $(3, 8)$, $(-2, -12)$

63. (g) $P(x) = P_O(x) + P_E(x)$, siendo
$P_O(x) = x^5 + 6x^3 - 2x$ y $P_E(x) = -x^2 + 5$

65.



(a) 26 licuadoras
(b) La utilidad máxima es \$3,276.22 cuando se producen 166 licuadoras

67. (a) Máximo local en $(1.8, 2.1)$,
mínimo local en $(3.5, -0.6)$
(b) Máximo local en $(1.8, 7.1)$, mínimo local en $(3.5, 4.4)$

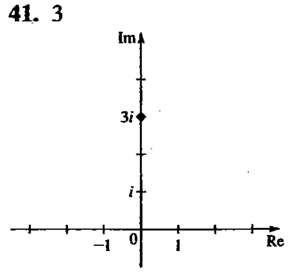
Sección 6.2

En las respuestas 1 a 11, el primer polinomio que aparece es el cociente, y el segundo es el residuo.

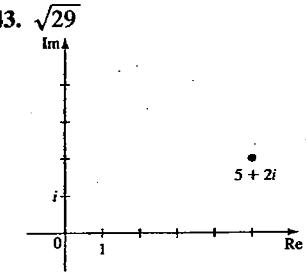
1. $x^2 + 2, -3$ **3.** $x^2 - 3x + 1, -1$
5. $x^4 + x^3 + 4x^2 + 4x + 4, -2$ **7.** $x + 2, 8x - 1$
9. $3x + 1, 7x - 5$ **11.** $2x^2 + 4x, 1$ **13.** -3
15. 12 **17.** -483 **19.** $\frac{7}{3}$ **21.** $\pm1, \pm3$
23. $\pm1, \pm2, \pm3, \pm6, \pm\frac{1}{2}, \pm\frac{3}{2}$ **25.** $-2, 2, 3$
27. $-1, 2, 3$ **29.** $1, 2, 4$ **31.** -1 **33.** $\pm1, \pm2$
35. $1, -1, -2, -4$ **37.** $\pm2, \pm\frac{3}{2}$ **39.** $-\frac{3}{2}, \frac{1}{2}, 1$
41. $-1, \pm\frac{1}{2}$ **43.** $-1, \pm2, 3$ **47.** $x^3 - 3x^2 - x + 3$
49. $x^4 - 8x^3 + 14x^2 + 8x - 15$
51. 1 positivas, 2 o 0 negativas; 3 o 1 reales
53. 0 positivas, 4, 2 o 0 negativas; 4, 2 o 0 reales
55. 5, 3 o 1 positivas, 2 o 0 negativas; 7, 5, 3 o 1 reales
61. $3, -2$ **63.** $3, -1$ **65.** $-2, \frac{1}{2}, \pm1$
67. $\pm\frac{1}{2}, \pm\sqrt{5}$ **69.** $-2, 1, 3, 4$ **71.** $-2, 2, 3$
73. $-\frac{3}{2}, -1, 1, 4$ **75.** 5.15 **77.** 0.67
79. $-1.00, 1.50$ **81.** -1.50 **83.** 11.3 pies
85. (a) Comenzó a nevar otra vez. **(b)** No
(c) Justo antes de la medianoche del sábado
87. 2.76 m **89.** 88 pulg (o 3.21 pulg)

Sección 6.3

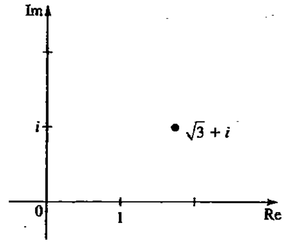
1. Parte real 3, parte imaginaria -5
3. Parte real 0, parte imaginaria 6
5. Parte real $\sqrt{2}$, parte imaginaria $\sqrt{3}$
7. $9 + i$ **9.** $12 + i$ **11.** $-19 + 4i$ **13.** $-4 + 8i$
15. $30 + 10i$ **17.** $-33 - 56i$ **19.** $-i$ **21.** $\frac{8}{5} + \frac{1}{5}i$
23. $-5 + 12i$ **25.** $-4 + 2i$ **27.** $-i$ **29.** 1
31. $5i$ **33.** -6 **35.** $(3 + \sqrt{5}) + (3 - \sqrt{5})i$ **37.** 2
39. $-i\sqrt{2}$
41. 3



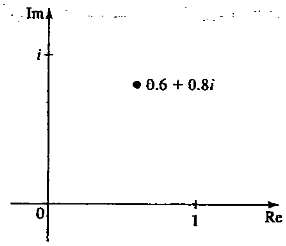
43. $\sqrt{29}$



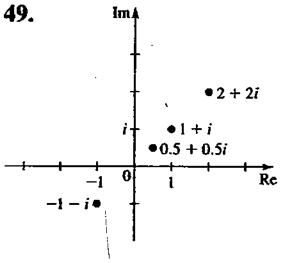
45. 2



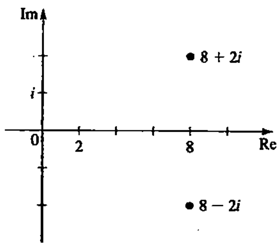
47. 1



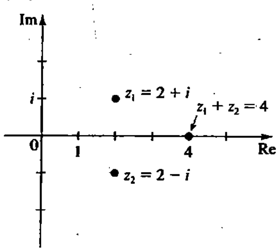
49.



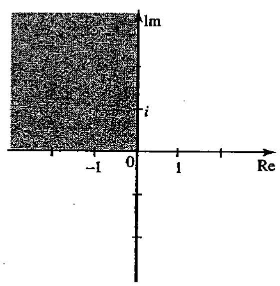
51.



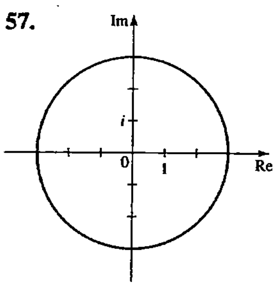
53.



55.



57.



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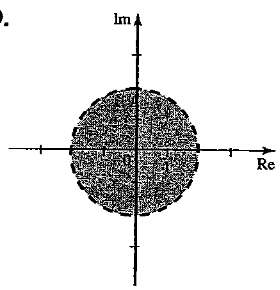
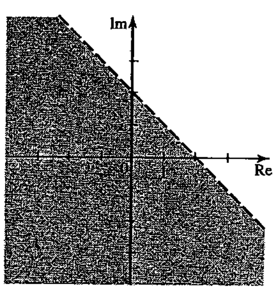


61.



Sección 6.4

1. $\pm 4i$ **3.** $-1 \pm i$ **5.** $-2 \pm 2i$ **7.** $\dfrac{5 \pm i\sqrt{23}}{6}$

9. $4 \pm i$ **11.** $\dfrac{-3 \pm i\sqrt{3}}{2}$ **13.** $\pm 1, \pm i$

15. $-2, 1 \pm i\sqrt{3}$ **17.** $\pm 2, \pm 2i$ **19.** $\pm 3, \dfrac{\pm 3 \pm 3\sqrt{3}\,i}{2}$

21. $\pm i\sqrt{5}$ **23.** $x^3 - 2x^2 + x - 2$

25. $x^4 - 4x^3 + 10x^2 - 12x + 5$

27. $6x^4 - 12x^3 + 18x^2 - 12x + 12$ **29.** $-2, \pm 2i$

31. $1, \dfrac{1 \pm i\sqrt{3}}{2}$ **33.** $2, \dfrac{1 \pm i\sqrt{3}}{2}$ **35.** $-2, 1, \pm 3i$

37. $(x + 3)\left(x - \dfrac{3 + 3\sqrt{3}\,i}{2}\right)\left(x - \dfrac{3 - 3\sqrt{3}\,i}{2}\right)$

39. $(x + 2)(x - 1)(x - 1 + i\sqrt{3})(x - 1 - i\sqrt{3}) \cdot \left(x + \dfrac{1 + i\sqrt{3}}{2}\right)\left(x + \dfrac{1 - i\sqrt{3}}{2}\right)$

41. $(x - 2)(x + 1 + i\sqrt{2})(x + 1 - i\sqrt{2})$

43. $(x - 2)(x + 1)(x - 3i)(x + 3i)$

45. 0 positivas, 2 o 0 negativas; 4 o 2 imaginarias

47. 1 positivas, 2 o 0 negativas; 4 o 2 imaginarias

49. **(a)** 4 reales **(b)** 2 reales, 2 imaginarias **(c)** 4 imaginarias

Sección 6.5

1. Intersección en x 6, intersección en y −6

3. Intersección en x 0, intersección en y 0

5. Intersección en x 3, intersección en y 3 vertical en $x = 2$, horizontal en $y = 2$

7. Vertical en $x = -3$, horizontal en $y = 0$

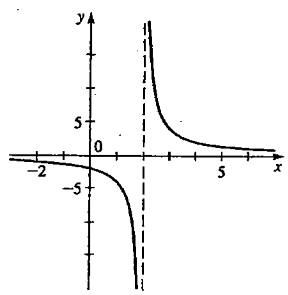
9. Vertical en $x = 3$ y $x = -2$; horizontal en $y = 1$

11. Horizontal en $y = 0$

13. Vertical en $x = 1$; inclinada $y = x + 1$

15. Vertical en $x = -3$

17.



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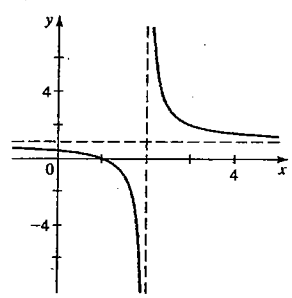
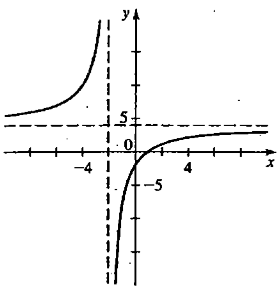


21.



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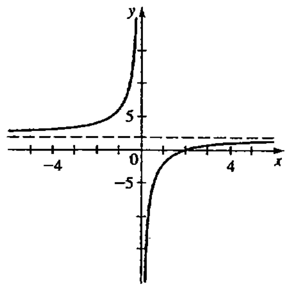
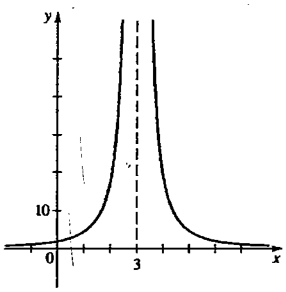
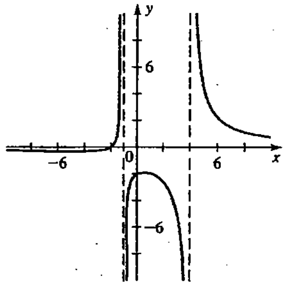


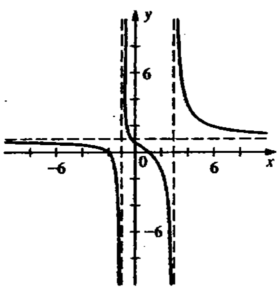
25.



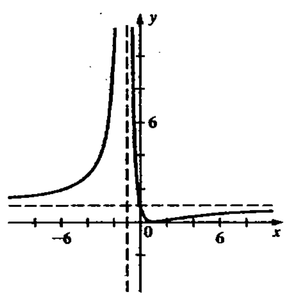
27.



29.



31.



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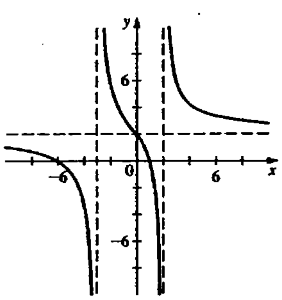

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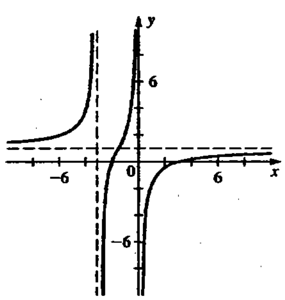

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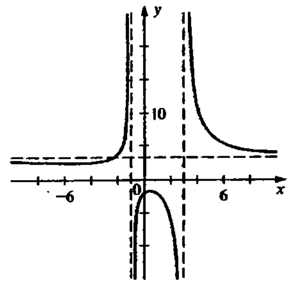

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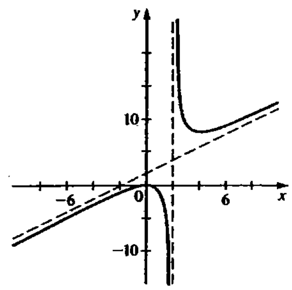
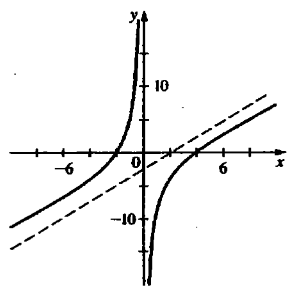
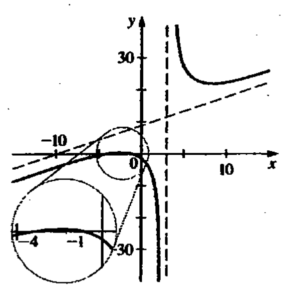


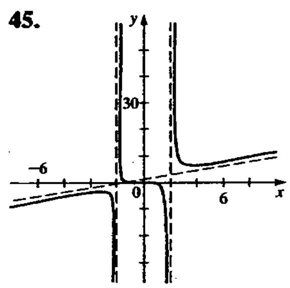
41.



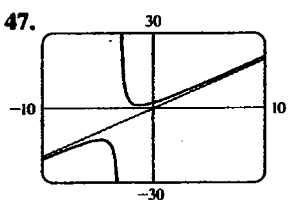
43.



45.

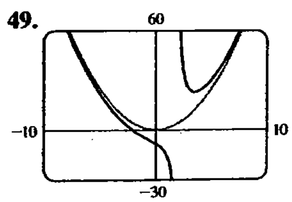


47.



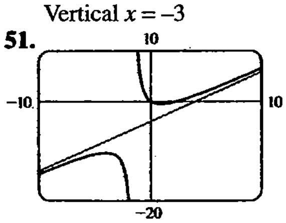
Vertical $x = -3$

49.



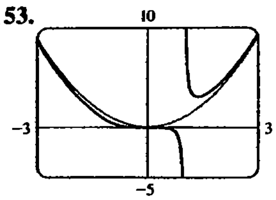
Vertical $x = 2$

51.



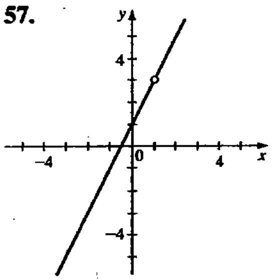
vertical $x = -1.5$
intersección en x 0, 2.5
intersección en y 0
máximo local $(-3.9, -10.4)$
mínimo local $(0.9, -0.6)$
comportamiento en extremos: $y = x - 4$

53.

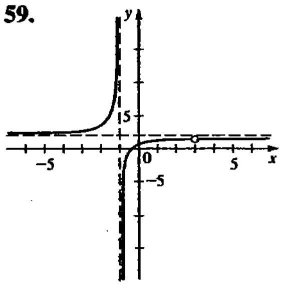


vertical $x = 1$
intersección en x 0
intersección en y 0
mínimo local (1.4, 3.1)
comportamiento en extremos: $y = x^2$

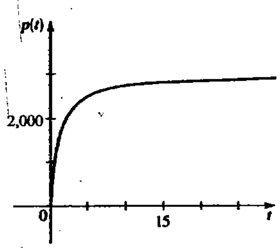
57.



59.



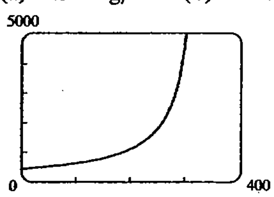
61. **(a)**



(b) Se nivela en 3,000

63. **(a)** 2.50 mg/L **(b)** Decrece a 0 **(c)** 16.61 h

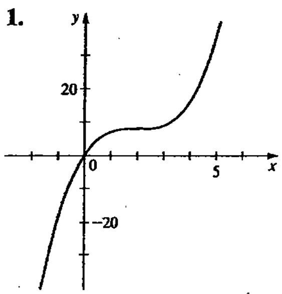
65.



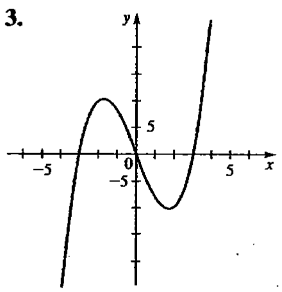
Si la rapidez del tren se acerca a la rapidez del sonido, el tono o altura aumenta indefinidamente (un estruendo sónico)

Repaso del capítulo 6

1.



3.



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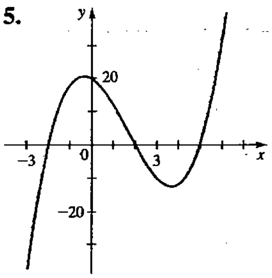
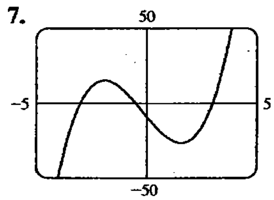
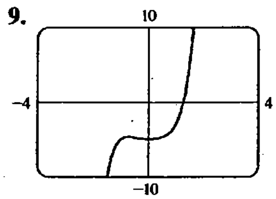


7.



intersecciones en x: −3, −0.5, 3
intersección en y: −9
máximo local: (−1.9, 15.1)
mínimo local: (1.6, −27.1)
$y \to \infty$ cuando $x \to \infty$
$y \to -\infty$ cuando $x \to -\infty$

9.



intersección en y: 1.3
intersección en y: −5
máximo local: (−0.7, −4.7)
mínimo local: (0, −5)
$y \to \infty$ cuando $x \to \infty$
$y \to -\infty$ cuando $x \to -\infty$

En las respuestas 11 a 17, el primer polinomio que aparece es el cociente y el segundo es el residuo.

11. $x^2 + 2x + 7$, 10 **13.** $x - 3$, -9

15. $x^3 - 5x^2 + 4$, -5

17. $x^3 + (\sqrt{3} + 1)x^2 + (\sqrt{3} + 1)x + \sqrt{3}$, 2

19. 3 **23.** 8 **25.** ±1, ±2, ±3, ±6, ±9, ±18

27. $-3 - 9i$ **29.** $19 + 40i$ **31.** $(-5 - 12i)/13$

33. i **35.** $(2 + 2\sqrt{3}) + (2 - 2\sqrt{3})i$

37. $4x^3 - 18x^2 + 14x + 12$

39. No; ya que los complejos conjugados de ceros imaginarios también son ceros, el polinomio tendría 8 ceros, contraviniendo el requisito que debe tener el grado 4.

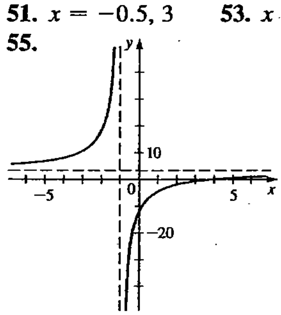
41. −3, 1, 5 **43.** $-1 \pm 2i$, −2 (multiplicidad 2)

45. 2, 1 (multiplicidad 3) **47.** $\pm 2, \pm 1 \pm i\sqrt{3}$

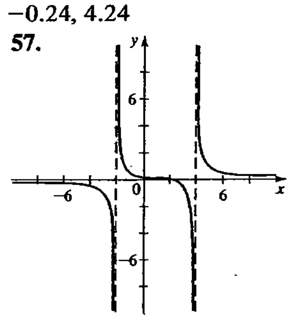
49. 1, 3, $\dfrac{-1 \pm i\sqrt{7}}{2}$

51. $x = -0.5, 3$ **53.** $x \approx -0.24, 4.24$

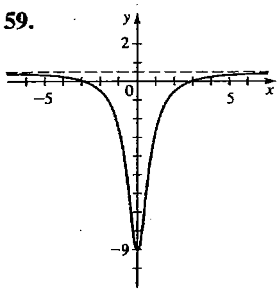
55.



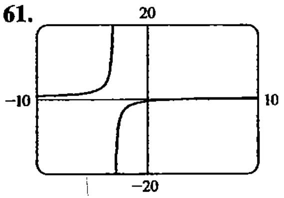
57.



59.

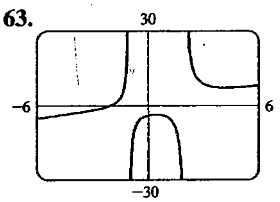


61.



abscisa al origen 3
ordenada al origen -0.5
vertical: $x = -3$
horizontal: $y = 0.5$
no hay extremos locales

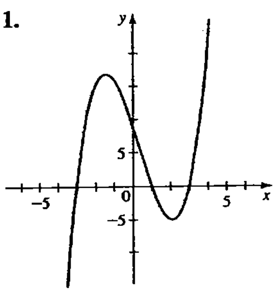
63.



intersección en x -2
intersección en y -4
vertical en $x = -1$, $x = 2$
inclinada $y = x + 1$
máximo local en $(0.425, -3.599)$
mínimo local en $(4.216, 7.175)$

Examen del capítulo 6

1.



2. cociente: $x^3 + 2x^2 + 2$
residuo: 9

3. **(a)** $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 3$
(b) $\left(x - \frac{1}{2}\right)(x - 1)(x + 1)(x - 3)$ **(c)** $-1, \frac{1}{2}, 1, 3$

4. **(a)** $\frac{6}{13} - \frac{22}{13}i$ **(b)** $2 - 11i$ **(c)** i

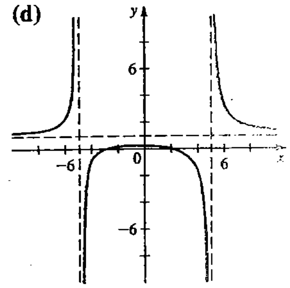
5. $-1, 2, -1 \pm i$

6. **(a)** Para P y Q: con el teorema de los ceros racionales; para R: con la regla de Descartes **(b)** No, según la regla de Descartes **(c)** Dos (una positiva y una negativa); con la regla de Descartes **(d)** Los únicos ceros racionales posibles son 1 y -1, y ninguno de ellos es cero

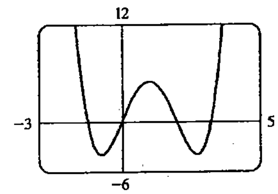
7. $x^4 + 2x^2 + 8x + 5$

8. **(a)** 4, 2 o 0 raíces reales positivas; 0 raíz real negativa

9. **(a)** r, u **(b)** s **(c)** s **(d)**



10.



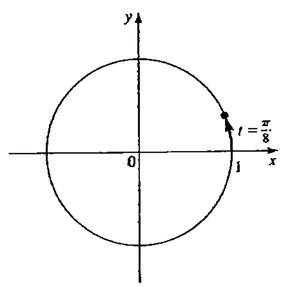
intersecciones en x: $-1.24, 0, 2, 3.24$
máximo local en $(1, 5)$
mínimos locales en $(-0.73, -4), (2.73, -4)$

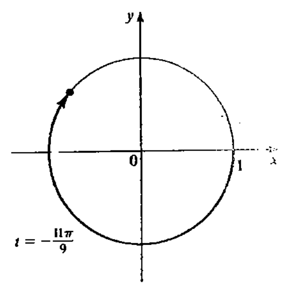
CAPÍTULO 7

Sección 7.1

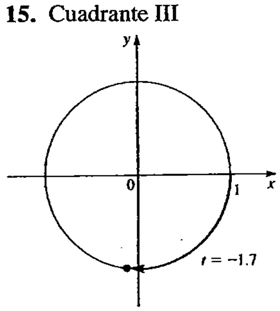
5. $P\left(\frac{3}{5}, \frac{4}{5}\right)$ **7.** $P\left(\frac{2}{3}, -\sqrt{5}/3\right)$ **9.** $P\left(\sqrt{2}/3, -\sqrt{7}/3\right)$

11. Cuadrante I **13.** Cuadrante II





15. Cuadrante III



17. $\left(\sqrt{2}/2, \sqrt{2}/2\right)$
19. $\left(\frac{1}{2}, -\sqrt{3}/2\right)$
21. $(-1, 0)$

23. $\left(-\frac{1}{2}, \sqrt{3}/2\right)$ **25.** $\left(-\sqrt{2}/2, -\sqrt{2}/2\right)$

27. **(a)** $\left(-\frac{3}{5}, \frac{4}{5}\right)$ **(b)** $\left(\frac{3}{5}, -\frac{4}{5}\right)$ **(c)** $\left(-\frac{3}{5}, -\frac{4}{5}\right)$ **(d)** $\left(\frac{3}{5}, \frac{4}{5}\right)$

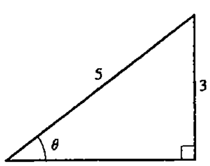
29. (a) $\pi/4$ (b) $\pi/3$ (c) $\pi/3$ (d) $\pi/6$
31. (a) $\pi/5$ (b) $\pi/6$ (c) $\pi/3$ (d) $\pi/6$
33. (a) $\pi/4$ (b) $(-\sqrt{2}/2, \sqrt{2}/2)$
35. (a) $\pi/3$ (b) $(-\frac{1}{2}, -\sqrt{3}/2)$
37. (a) $\pi/4$ (b) $(-\sqrt{2}/2, -\sqrt{2}/2)$
39. (a) $\pi/6$ (b) $(-\sqrt{3}/2, -\frac{1}{2})$
41. (a) $\pi/3$ (b) $(\frac{1}{2}, \sqrt{3}/2)$
43. (a) $\pi/3$ (b) $(-\frac{1}{2}, -\sqrt{3}/2)$ **45.** $(0.5, 0.8)$
47. $(0.5, -0.9)$

Sección 7.2 ■

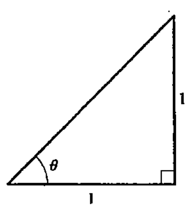
1. $2\pi/9 \approx 0.698$ rad **3.** $2\pi/5 \approx 1.257$ rad
5. $\pi/4 \approx 0.785$ rad
7. $-630°$
9. $360°/\pi \approx 114.6°$ **11.** $40°$ **17.** $36°$
15. $660°$, $1020°$, $-60°$, $-420°$
13. $7\pi/4$, $15\pi/4$, $-9\pi/4$, $-17\pi/4$
17. Sí **19.** Sí
21. $63°$ **23.** $280°$
25. π **41.** $\pi/4$ **27.** $55\pi/9 \approx 19.2$ **29.** 4
31. 4 mi **33.** 2 rad $\approx 114.6°$ **35.** $36/\pi \approx 11.459$ m
37. $330\pi \approx 1037$ mi **39.** 1.6 millones de millas **41.** 1.15 mi
43. 50 m^2 **45.** 4 m **47.** 6 cm^2

Sección 7.3 ■

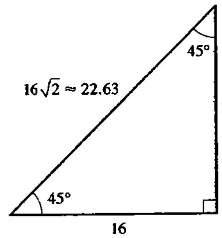
1. $\operatorname{sen}\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$, $\tan\theta = \frac{4}{3}$, $\csc\theta = \frac{5}{4}$, $\sec\theta = \frac{5}{3}$, $\cot\theta = \frac{3}{4}$
3. $\operatorname{sen}\theta = \frac{4}{7}$, $\cos\theta = \sqrt{33}/7$, $\tan\theta = 4\sqrt{33}/33$, $\csc\theta = \frac{7}{4}$, $\sec\theta = 7\sqrt{33}/33$, $\cot\theta = \sqrt{33}/4$
5. (a) $2\sqrt{13}/13, 2\sqrt{13}/13$ (b) $\frac{2}{3}, \frac{2}{3}$ (c) $\sqrt{13}/3, \sqrt{13}/3$
7. $\frac{25}{2}$ **9.** $13\sqrt{3}/2$ **11.** 16.51658
13. $x = 28\cos\theta$, $y = 28\operatorname{sen}\theta$
15. $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$, $\csc\theta = \frac{5}{3}$, $\sec\theta = \frac{5}{4}$, $\cot\theta = \frac{4}{3}$



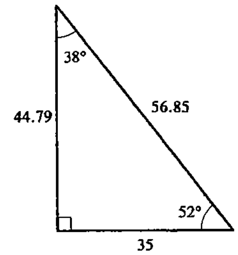
17. $\operatorname{sen}\theta = \sqrt{2}/2$, $\cos\theta = \sqrt{2}/2$, $\tan\theta = 1$, $\csc\theta = \sqrt{2}$, $\sec\theta = \sqrt{2}$



21. $(1 + \sqrt{3})/2$ **23.** 1 **25.** $\frac{1}{2}$
27.



29.



31. $\operatorname{sen}\theta \approx 0.45$, $\cos\theta \approx 0.89$, $\tan\theta \approx 0.50$, $\csc\theta \approx 2.24$, $\sec\theta \approx 1.12$, $\cot\theta \approx 2.00$
33. 1,026 pies **35.** (a) 2,100 mi (b) No **37.** 19 pies
39. 38.7° **41.** 345 pies **43.** 415 pies, 152 pies
45. 2,570 pies **47.** 5,808 pies **49.** 91.7 millones de mi
51. 3,960 mi **53.** 230.9 **55.** 63.7
57. $a = \operatorname{sen}\theta$, $b = \tan\theta$, $c = \sec\theta$, $d = \cos\theta$

Sección 7.4 ■

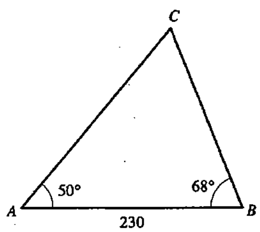
1. (a) 0 (b) 1 **3.** (a) 0 (b) -1
5. (a) 1 (b) -1 **7.** (a) 0 (b) 0
9. (a) $\frac{1}{2}$ (b) 2 **11.** (a) $\frac{1}{2}$ (b) $\frac{1}{2}$
13. (a) $\sqrt{3}/3$ (b) $-\sqrt{3}/3$ **15.** (a) 2 (b) $-2\sqrt{3}/3$
17. $\operatorname{sen} 0 = 0$, $\cos 0 = 1$, $\tan 0 = 0$, $\sec 0 = 1$, los otros están indefinidos
19. $\operatorname{sen}\pi = 0$, $\cos\pi = -1$, $\tan\pi = 0$, $\sec\pi = -1$, los otros están indefinidos
21. $\frac{4}{5}, \frac{3}{5}, \frac{4}{3}$ **23.** $-\sqrt{13}/7, \frac{6}{7}, -\sqrt{13}/6$ **25.** $\frac{9}{41}, \frac{40}{41}, \frac{9}{40}$
27. $-\frac{12}{13}, -\frac{5}{13}, \frac{12}{5}$ **29.** (a) 0.8 (b) 0.84147
31. (a) 0.9 (b) 0.93204 **33.** (a) 1 (b) 1.02964
35. (a) -0.6 (b) -0.57482 **37.** Positivo
39. Negativo **41.** II **43.** II
45. $\operatorname{sen} t = \sqrt{1 - \cos^2 t}$
47. sen t = - $\sqrt{1\text{-}1/\sec^2 t}$ = - $\sqrt{\sec^2 t - 1 / \sec t}$.
49. $\sec^2 t \operatorname{sen}^2 t = 1 - \cos^2 t / \cos^2 t = 1/\cos^2 t - 1$.
51. sen t = -3/5, tan t = ¾sec t = -5/4, cosc t = -5/3.
53. cos t = 1/3, sen t = -2/3 √2, tan t = - 2 √2, cot t = -√2 /4, csc t = -3/4 √2.
55. sen t = -√221/17, cos t = -4 √17/ 17, cot t = 4, sec t = - √17/ 4, csc t = √221/12.
57. sen t = 4 √17/17, cos t = – √221/17, cot t = -1/4, sec t = - √221/13, cos t = √17/4.
59. Par.
61. Ninguna.
63. Par.
65. Par.

Sección 7.5

1. **(a)** 45° **(b)** 35° **(c)** 1°
3. **(a)** 25° **(b)** 85° **(c)** 15°
5. **(a)** $\pi/3$ **(b)** $\pi/4$ **(c)** $\pi - 3 \approx 0.14$
7. $\frac{1}{2}$ **9.** $\sqrt{2}/2$ **11.** $-\sqrt{3}/2$ **13.** 1 **15.** $-\sqrt{3}/2$
17. $\sqrt{3}/3$ **19.** $\sqrt{3}/2$ **21.** -1 **23.** $\frac{1}{2}$ **25.** 2
27. -1 **29.** indefinido **31.** III **33.** IV
35. $\tan\theta = -\sqrt{1-\cos^2\theta}/\cos\theta$ **37.** $\cos\theta = \sqrt{1-\operatorname{sen}^2\theta}$
39. $\sec\theta = -\sqrt{1+\tan^2\theta}$
41. $\cos\theta = -\frac{4}{5}$, $\tan\theta = -\frac{3}{4}$, $\csc\theta = \frac{5}{3}$, $\sec\theta = -\frac{5}{4}$, $\cot\theta = -\frac{4}{3}$
43. $\operatorname{sen}\theta = -\frac{3}{5}$, $\cos\theta = \frac{4}{5}$, $\csc\theta = -\frac{5}{3}$, $\sec\theta = \frac{5}{4}$, $\cot\theta = -\frac{4}{3}$
45. $\operatorname{sen}\theta = \frac{1}{2}$, $\cos\theta = \sqrt{3}/2$, $\tan\theta = \sqrt{3}/3$, $\sec\theta = 2\sqrt{3}/3$, $\cot\theta = \sqrt{3}$
47. $\operatorname{sen}\theta = 3\sqrt{5}/7$, $\tan\theta = -3\sqrt{5}/2$, $\csc\theta = 7\sqrt{5}/15$, $\sec\theta = -\frac{7}{2}$, $\cot\theta = -2\sqrt{5}/15$
49. **(a)** $\sqrt{3}/2$, $\sqrt{3}$ **(b)** $\frac{1}{2}$, $\sqrt{3}/4$ **(c)** $\frac{3}{4}$, 0.88967
51. 19.1 **53.** 66.1°

Sección 7.6

1. 21.5 **3.** 134.6



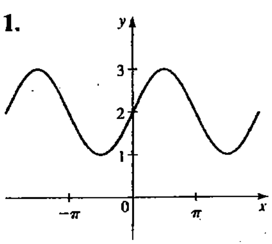
15. $\angle B \approx 30°$, $\angle C \approx 40°$, $c \approx 19$ **17.** No hay solución
19. $\angle A_1 \approx 125°$, $\angle C_1 \approx 30°$, $a_1 \approx 49$; $\angle A_2 \approx 5°$, $\angle C_2 \approx 150°$, $a_2 \approx 5.6$
21. No hay solución **23.** 219 pies
25. **(a)** 1018 mi **(b)** 1017 mi **27.** 155 m
29. **(a)** $d\dfrac{\operatorname{sen}\alpha}{\operatorname{sen}(\beta-\alpha)}$ **(c)** 2,350 pies **31.** 48.2°

Sección 7.7

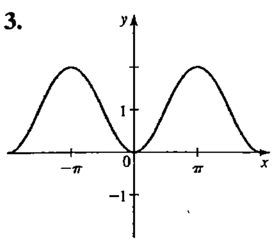
1. 13 **3.** 29.89°
5. $\angle A \approx 39.4°$, $\angle B \approx 20.6°$, $c \approx 24.6$
7. a = 57,15, B ≈ 80° y C ≈ 30°.
9. A ≈ 38°, B ≈ 49° y C ≈ 123°.

Sección 7.8

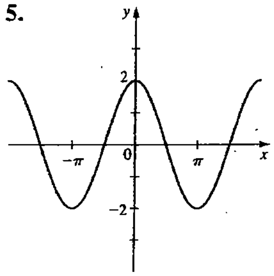
1.



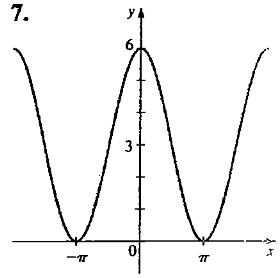
3.



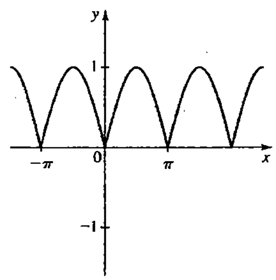
5.



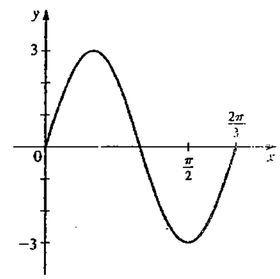
7.



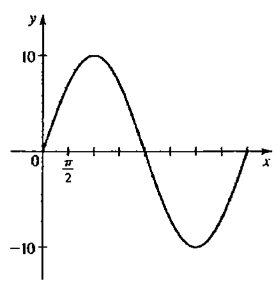
9.



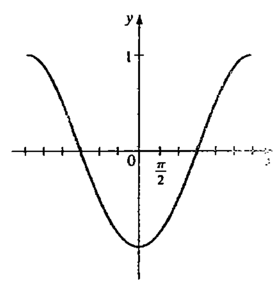
11. 3, $2\pi/3$



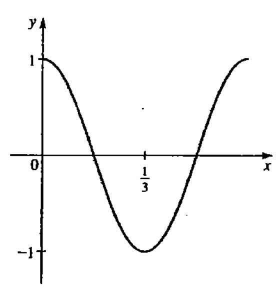
13. 10, 4π



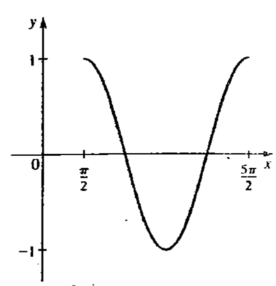
15. 1, 6π



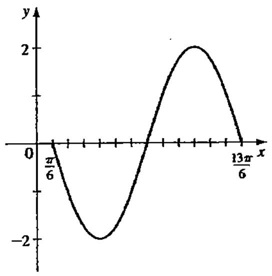
17. 3, $\frac{2}{3}$



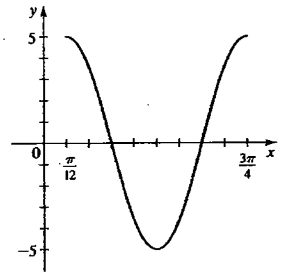
19. 1, 2π, $\pi/2$



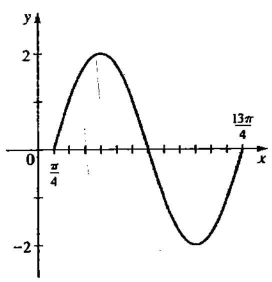
21. $2, 2\pi, \pi/6$



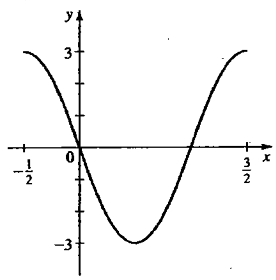
23. $5, 2\pi/3, \pi/12$



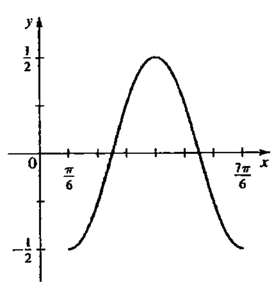
25. $2, 3\pi, \pi/4$



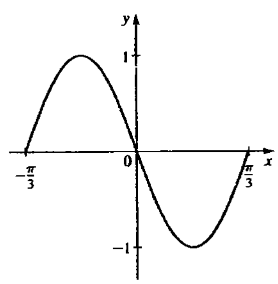
27. $3, 2, -\frac{1}{2}$



29. $\frac{1}{2}, \pi, \pi/6$

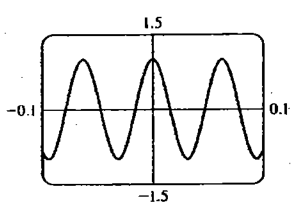


31. $1, 2\pi/3, -\pi/3$

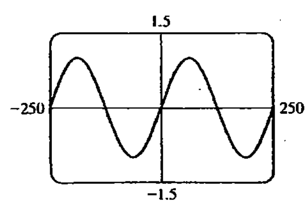


33. **(a)** $4, 2\pi, 0$ **(b)** $y = 4\,\text{sen}\,x$
35. **(a)** $3, 4\pi, 0$ **(b)** $y = 3\,\text{sen}\,\frac{1}{2}x$
37. **(a)** $\frac{1}{2}, \pi, -\pi/3$ **(b)** $y = -\frac{1}{2}\cos 2(x + \pi/3)$

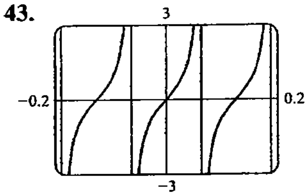
39.



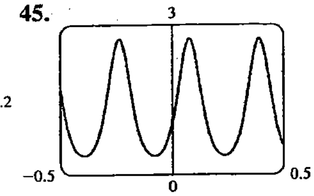
41.



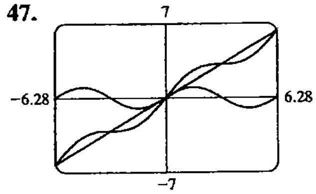
43.



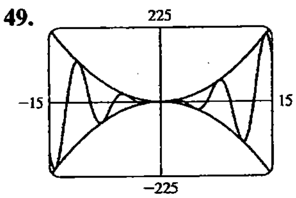
45.



47.

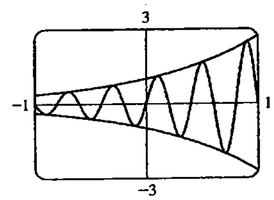


49.



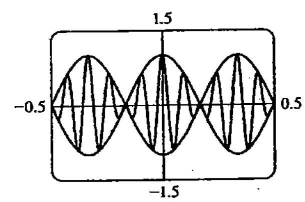
$y = x^2$ sen x es una curva seno que está entre las gráficas de $y = x^2$ y $y = -x^2$

51.



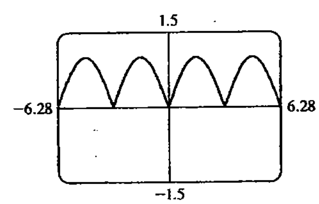
$y = e^x$ sen $5\pi x$ es una curva seno que está entre las gráficas de $y = e^x$ y $y = e^{-x}$

53.



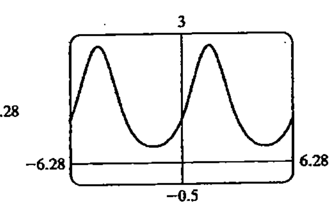
$y = \cos 3\pi x \cos 21\pi x$ es una curva coseno que está entre las gráficas de $y = \cos 3\pi x$ y $y = -\cos 3\pi x$

55. **(a)**



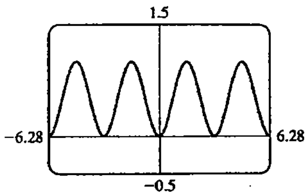
(b) Periodo π **(c)** Par

57. **(a)**



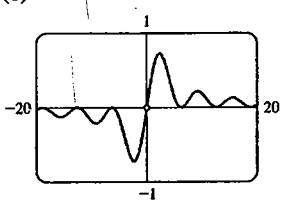
(b) Periodo 2π **(c)** Par

59. **(a)**



(b) Periodo π
(c) Par

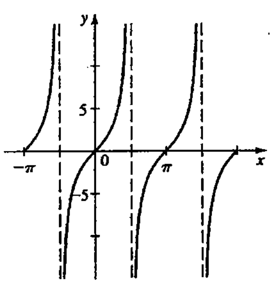
61. Valor máximo 1.76 cuando x 0.94; valor mínimo −1.76 cuando x −0.94 (Los mismos valores máximos y mínimos se presentan en una cantidad infinita de otros valores de x.)
63. Valor máximo 3.00 cuando $x = 1.57$, valor mínimo −1.00 cuando $x = -1.57$ (Los mismos valores máximos y mínimos ocurren en un número infinito de otros valores de x.)
65. 1.16 **67.** 0.34, 2.80
69. **(a)** Impar **(b)** 0, $\pm 2\pi$, $\pm 4\pi$, $\pm 6\pi, \ldots$
(c)



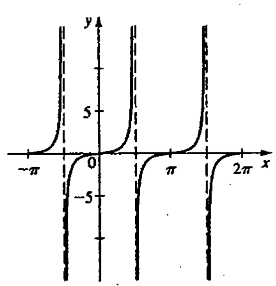
(d) $f(x)$ tiende a 0
(e) $f(x)$ tiende a 0

Sección 7.9 ■

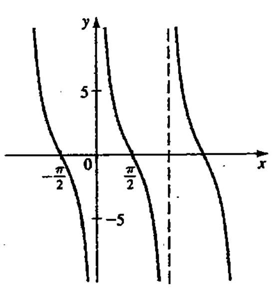
1. π



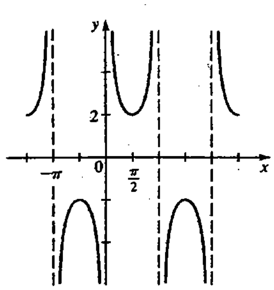
3. π



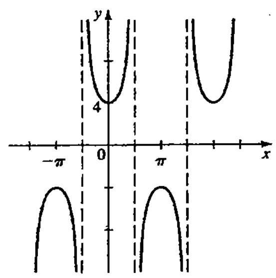
5. π



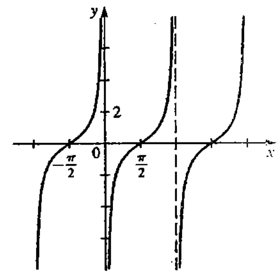
7. 2π



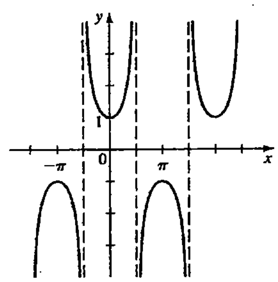
9. 2π



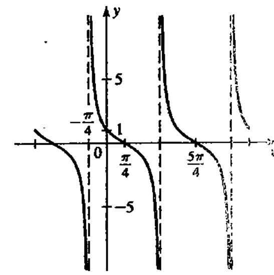
11. π



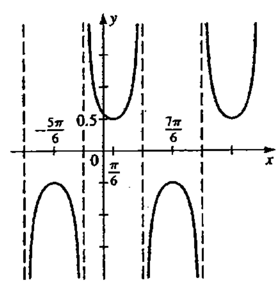
13. 2π



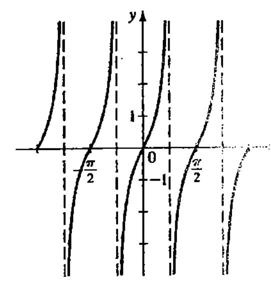
15. π



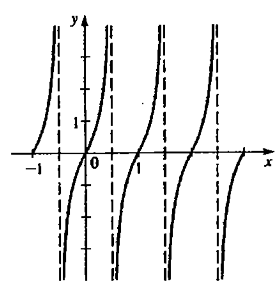
17. 2π



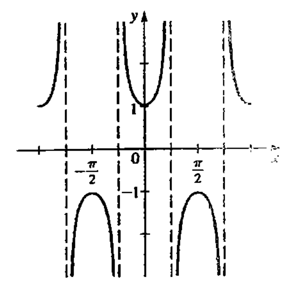
19. $\pi/2$



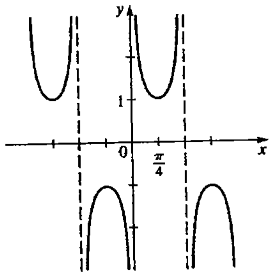
21. 1



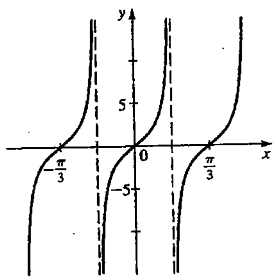
23. π



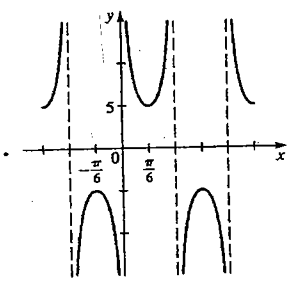
25. π



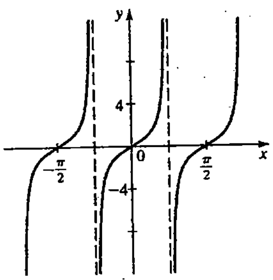
27. $\pi/3$



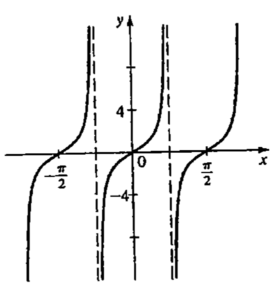
29. $2\pi/3$



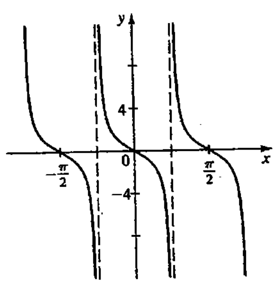
31. $\pi/2$



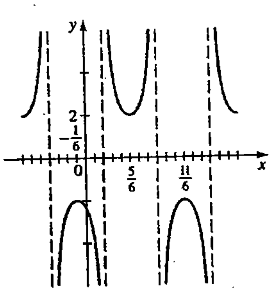
33. $\pi/2$



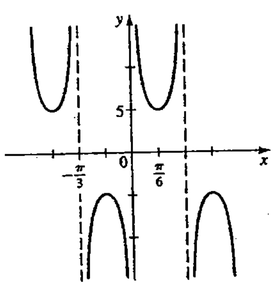
35. $\pi/2$



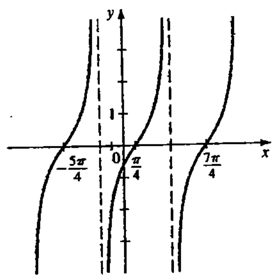
37. 2



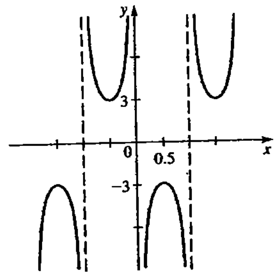
39. $2\pi/3$



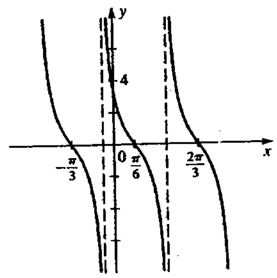
41. $3\pi/2$



43. 2

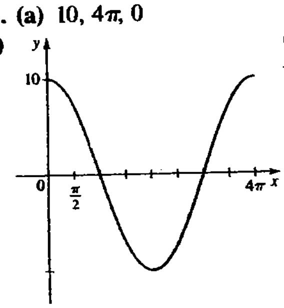


45. $\pi/2$

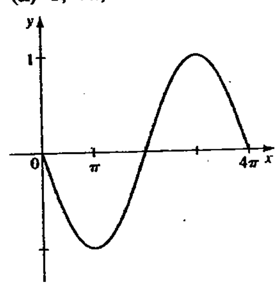


Repaso del capítulo 7 ■

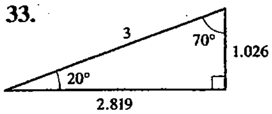
1. (b) $\sqrt{3}/2, \frac{1}{2}, \sqrt{3}$
3. (a) $\pi/3$ **(b)** $\left(\frac{1}{2}, \sqrt{3}/2\right)$ **(c)** $\operatorname{sen} t = \sqrt{3}/2, \cos t = \frac{1}{2}$, $\tan t = \sqrt{3}$, $\csc t = 2\sqrt{3}/3$, $\sec t = 2$, $\cot t = \sqrt{3}/3$
5. (a) 0.89121 **(b)** 0.45360
7. $(\operatorname{sen} t)/(1 - \operatorname{sen}^2 t)$
9. $\tan t = -\frac{5}{12}$, $\csc t = \frac{13}{5}$, $\sec t = -\frac{13}{12}$, $\cot t = -\frac{12}{5}$
11. $(16 - \sqrt{17})/4$
13. (a) 10, 4π, 0
(b)



15. (a) 1, 4π, 0
(b)



17. $y = 5 \operatorname{sen} 4x$
19. (a) $7\pi/18 \approx 1.22$ **(b)** $7\pi/3 \approx 7.33$ **(c)** $-4\pi/3 \approx -4.19$ **(d)** $-2\pi/9 \approx -0.70$
21. 8 m **23.** 82 pies **25.** 0.619 rad $\approx 35.4°$
27. 18,151 pies^2
29. $\operatorname{sen} \theta = 5/\sqrt{74}$, $\cos \theta = 7/\sqrt{74}$, $\tan \theta = \frac{5}{7}$, $\csc \theta = \sqrt{74}/5$, $\sec \theta = \sqrt{74}/7$, $\cot \theta = \frac{7}{5}$
31. $x \approx 3.83$, $y \approx 3.21$

33. 

35. 48 m 37. 1,076 mi
39. $-\sqrt{2}/2$ 41. 1

43. $-\sqrt{3}/3$ 33. $-\sqrt{2}/2$

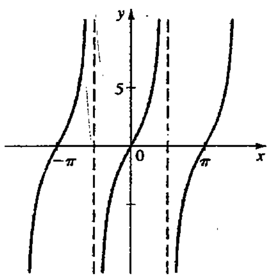
45. $\operatorname{sen}\theta = \frac{12}{13}$, $\cos\theta = -\frac{5}{13}$, $\tan\theta = -\frac{12}{5}$, $\csc\theta = \frac{13}{12}$, $\sec\theta = -\frac{13}{5}$, $\cot\theta = -\frac{5}{12}$

47. 60° 49. $\tan\theta = -\sqrt{1-\cos^2\theta}/\cos\theta$

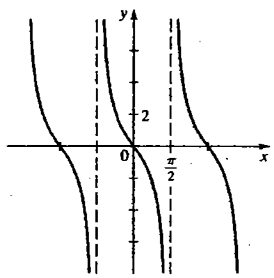
51. $\operatorname{sen}\theta = \sqrt{7}/4$, $\cos\theta = \frac{3}{4}$, $\csc\theta = 4\sqrt{7}/7$, $\cot\theta = 3\sqrt{7}/7$

53. $-\sqrt{5}/5$ 55. 1 57. 5.32

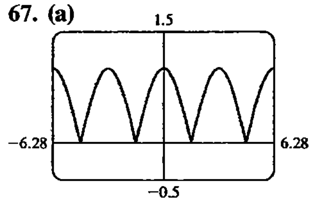
63. π



65. π

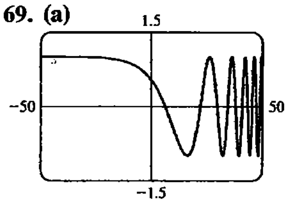


67. (a)



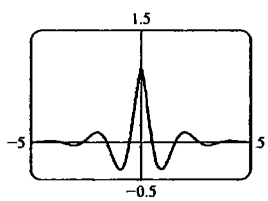
(b) Periodo π
(c) Par

69. (a)



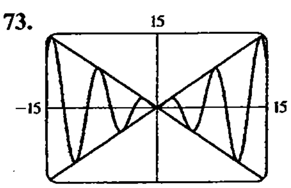
(b) No es periódica
(c) Ninguna de las dos

71. (a)



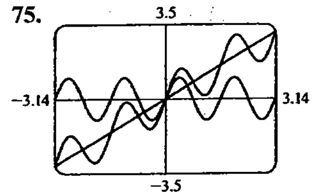
(b) No es periódica
(c) Par

73.



$y = x \operatorname{sen} x$ es una función seno cuya gráfica está entre las de $y = x$ y $y = -x$.

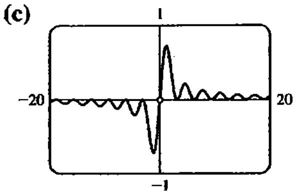
75.



Las gráficas se relacionan mediante suma.

77. 1.76, −1.76 79. 0.30, 2.84

81. (a) Impar (b) 0, $\pm\pi$, $\pm 2\pi$, ...
(c)



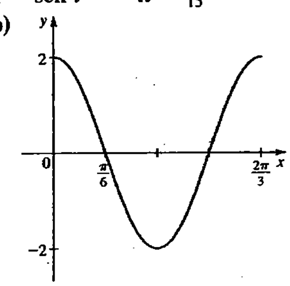
(d) $f(x)$ tiende a 0
(e) $f(x)$ tiende a 0

Examen del capítulo 7 ■

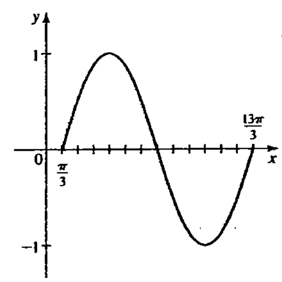
1. $y = -\frac{5}{13}$ 2. (a) $\frac{4}{5}$ (b) $-\frac{3}{5}$ (c) $\frac{4}{3}$ (d) $\frac{5}{3}$

3. $\tan t = -(\operatorname{sen} t)/\sqrt{1-\operatorname{sen}^2 t}$ 4. $-\frac{2}{15}$

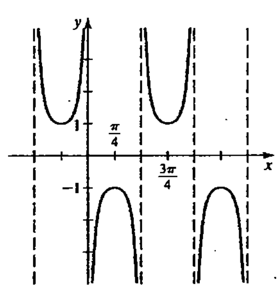
5. (a) 2, $2\pi/3$, 0 (b)



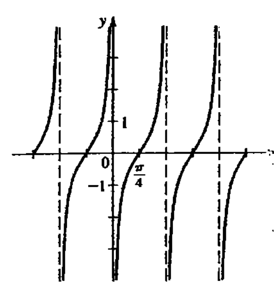
6. (a) 1, 4π, $\pi/3$



7. π

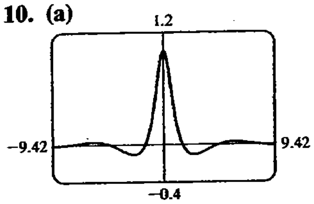


8. $\pi/2$



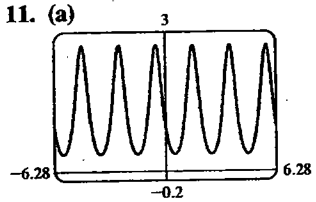
9. $y = 2\,\mathrm{sen}\,2(x + \pi/3)$

10. (a)



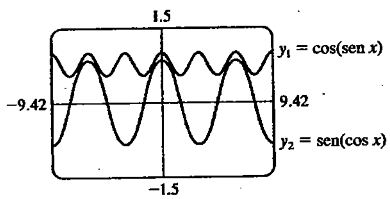
(b) Par
(c) Valor mínimo −0.11 cuando x 2.54, valor máximo 1 cuando $x = 0$

11. (a)



(b) Ninguna de las dos
(c) Valor mínimo 0.37, valor máximo 2.72 (Ambos valores se alcanzan en una cantidad infinita de valores de x.)

12. (a)



(b) y_1 tiene el periodo π, y_2 tiene el periodo 2π
(c) $\mathrm{sen}(\cos x) < \cos(\mathrm{sen}\, x)$

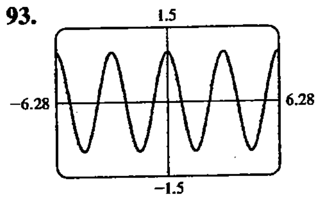
13. $5\pi/3, -\pi/10$ **14.** 150°, 137.5°
15. (a) $\sqrt{2}/2$ **(b)** $\sqrt{3}/3$ **(c)** 2 **(d)** 1
16. $(26 + 6\sqrt{13})/39$ **17.** $a = 24\,\mathrm{sen}\theta$, $b = 24\cos\theta$
18. $(4 - 3\sqrt{2})/4$ **7.** $-\frac{13}{12}$ **19.** $\tan\theta = -\sqrt{\sec^2\theta - 1}$
20. 19.6 pies **21.** 9.1 **22.** 250.5 **23.** 8.4 **24.** 19.5
25. 15.3 m^2 **26.** 24.3 m **27.** 129.9° **28.** 44.9
29. 554 pies

CAPÍTULO 8

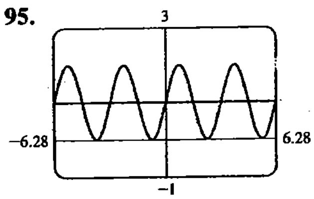
Sección 8.1 ■

1. $\mathrm{sen}\,x$ **3.** 1 **5.** $\sec u$ **7.** $\tan x$ **9.** $\mathrm{sen}\,y$
11. $\mathrm{sen}^2 x$ **13.** $\sec x$ **15.** $2\sec u$ **17.** $\cos^2 x$
19. $\cos\theta$ **83.** $\tan\theta$ **85.** $\tan\theta$ **87.** $3\cos\theta$

93. Sí

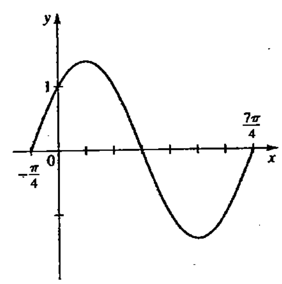


95. No



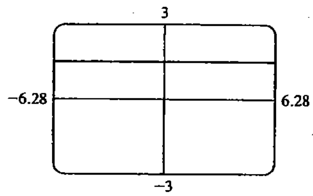
Sección 8.2 ■

1. $(\sqrt{6} - \sqrt{2})/4$ **3.** $-2 - \sqrt{3}$ **5.** $(\sqrt{6} - \sqrt{2})/4$
7. $\sqrt{2}/2$ **9.** $\sqrt{3}$ **33.** $2\,\mathrm{sen}\left(x + \dfrac{5\pi}{6}\right)$
35. $5\sqrt{2}\,\mathrm{sen}\left(2x + \dfrac{7\pi}{4}\right)$
37. $f(x) = \sqrt{2}\,\mathrm{sen}\left(x + \dfrac{\pi}{4}\right)$



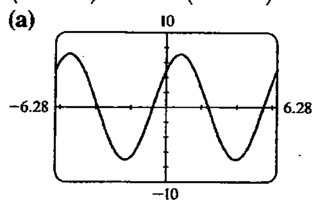
41. $\tan\gamma = \frac{17}{6}$

43. **(a)**



$\operatorname{sen}^2\left(x + \dfrac{\pi}{4}\right) + \operatorname{sen}^2\left(x - \dfrac{\pi}{4}\right) = 1$

45. **(a)**



(b) $k = 5\sqrt{2}$, $\theta = \pi/4$

Sección 8.3 ■

1. $\frac{120}{169}, \frac{119}{169}, \frac{120}{119}$ **3.** $-\frac{24}{25}, -\frac{7}{25}, \frac{24}{7}$ **5.** $\frac{24}{25}, \frac{7}{25}, \frac{24}{7}$

7. $\frac{1}{2}\left(\frac{3}{4} - \cos 2x + \frac{1}{4}\cos 4x\right)$

9. $\frac{1}{32}\left(\frac{3}{4} - \cos 4x + \frac{1}{4}\cos 8x\right)$

11. $\frac{1}{16}(1 - \cos 2x - \cos 4x + \cos 2x \cos 4x)$

13. $\frac{1}{2}\sqrt{2 - \sqrt{3}}$ **15.** $\frac{1}{2}\sqrt{2 + \sqrt{2}}$ **17.** $\frac{1}{2}\sqrt{2 - \sqrt{3}}$

19. **(a)** $\operatorname{sen} 36°$ **(b)** $\operatorname{sen} 6\theta$

21. **(a)** $\cos 68°$ **(b)** $\cos 10\theta$ **23.** **(a)** $\tan 4°$ **(b)** $\tan 2\theta$

25. $\sqrt{10}/10, 3\sqrt{10}/10, \frac{1}{3}$

27. $\sqrt{(3 + 2\sqrt{2})/6}, \sqrt{(3 - 2\sqrt{2})/6}, 3 + 2\sqrt{2}$

29. $\sqrt{6}/6, -\sqrt{30}/6, -\sqrt{5}/5$

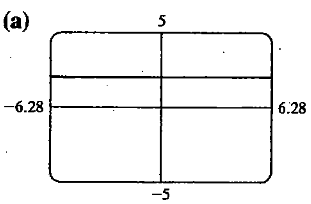
31. $\frac{1}{2}(\operatorname{sen} 5x - \operatorname{sen} x)$ **33.** $\frac{3}{2}(\cos 11x + \cos 3x)$

35. $2 \operatorname{sen} 4x \cos x$ **37.** $2 \operatorname{sen} 5x \operatorname{sen} x$

39. $-2\cos\frac{9}{2}x \operatorname{sen}\frac{5}{2}x$ **41.** $(\sqrt{2} + \sqrt{3})/2$

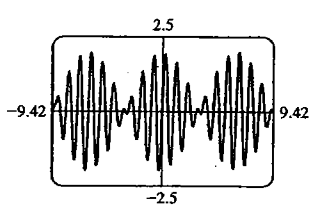
43. $\frac{1}{4}(\sqrt{2} - 1)$ **45.** $\sqrt{2}/2$

69. **(a)**

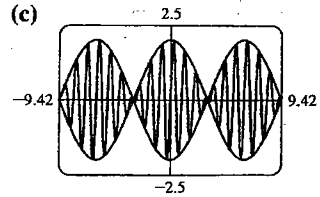


$\dfrac{\operatorname{sen} 3x}{\operatorname{sen} x} - \dfrac{\cos 3x}{\cos x} = 2$

71. **(a)**



(c)

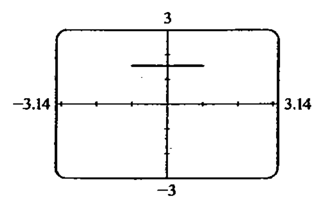


La gráfica de $y = f(x)$ está entre las otras dos gráficas.

75. **(a)** $P(t) = 8t^4 - 8t^2 + 1$
(b) $Q(t) = 16t^5 - 20t^3 + 5t$

Sección 8.4 ■

1. **(a)** $\pi/6$ **(b)** $\pi/3$ **(c)** No definida
3. **(a)** $\pi/4$ **(b)** $\pi/4$ **(c)** $-\pi/4$
5. **(a)** $\pi/2$ **(b)** 0 **(c)** π
7. **(a)** $\pi/6$ **(b)** $-\pi/6$ **(c)** No definida
9. **(a)** 0.87696 **(b)** 2.09601 **11.** $\frac{1}{3}$ **13.** 10
15. $\pi/3$ **17.** $-\pi/6$ **19.** $-\pi/3$ **21.** $\sqrt{3}/3$ **23.** $\frac{1}{2}$
25. $\pi/3$ **27.** $\frac{4}{5}$ **29.** $\frac{12}{13}$ **31.** $\frac{13}{5}$ **33.** $\sqrt{5}/5$
35. $\frac{24}{25}$ **37.** 1 **39.** $\sqrt{1 - x^2}$ **41.** $x/\sqrt{1 - x^2}$
43. $\dfrac{1 - x^2}{1 + x^2}$ **45.** 0 **47.** $\theta = \tan^{-1}\dfrac{50}{s}$
49. **(b)** 17.1°, 29.7°, 19.7°
51. **(a)**



Conjetura: $y = \pi/2$ para $-1 \leq x \leq 1$
53. **(a)** 0.28 **(b)** $(-3 + \sqrt{17})/4$

Sección 8.5 ■

1. $\dfrac{\pi}{3} + 2k\pi, \dfrac{5\pi}{3} + 2k\pi$ **3.** $\dfrac{\pi}{3} + 2k\pi, \dfrac{2\pi}{3} + 2k\pi$

5. $\dfrac{\pi}{3} + 2k\pi, \dfrac{2\pi}{3} + 2k\pi, \dfrac{4\pi}{3} + 2k\pi, \dfrac{5\pi}{3} + 2k\pi$

7. $\dfrac{(2k + 1)\pi}{4}$ **9.** $\dfrac{\pi}{2} + k\pi, \dfrac{7\pi}{6} + 2k\pi, \dfrac{11\pi}{6} + 2k\pi$

11. $-\dfrac{\pi}{3} + k\pi$ **13.** $\dfrac{\pi}{2} + k\pi$

15. $\dfrac{\pi}{3} + 2k\pi, \dfrac{5\pi}{3} + 2k\pi$ **17.** $\dfrac{3\pi}{2} + 2k\pi$

19. No hay solución **21.** $\dfrac{\pi}{18} + \dfrac{2k\pi}{3}, \dfrac{5\pi}{18} + \dfrac{2k\pi}{3}$

23. $4k\pi$ **25.** $\dfrac{k\pi}{3}$

27. $\frac{\pi}{6} + 2k\pi, \frac{2\pi}{3} + 2k\pi, \frac{5\pi}{6} + 2k\pi, \frac{4\pi}{3} + 2k\pi$

29. $\frac{\pi}{8} + \frac{k\pi}{2}, \frac{3\pi}{8} + \frac{k\pi}{2}$

31. $\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}$

33. $\frac{\pi}{6}, \frac{3\pi}{4}, \frac{5\pi}{6}, \frac{7\pi}{4}$

35. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

37. $0, \frac{2\pi}{3}, \frac{4\pi}{3}$

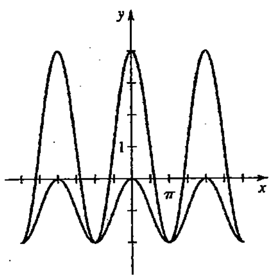
39. **(a)** 1.15928, 5.12391
(b) $1.15928 + 2k\pi, 5.12391 + 2k\pi$

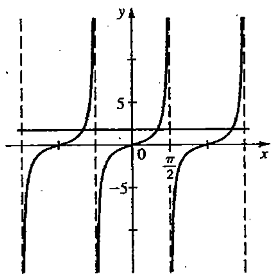
41. **(a)** 1.36944, 4.91375
(b) $1.36944 + 2k\pi, 4.91375 + 2k\pi$

43. **(a)** 0.46365, 2.67795, 3.60524, 5.81954
(b) $0.46365 + k\pi, 2.67795 + k\pi$

45. **(a)** 0.33984, 2.80176
(b) $0.33984 + 2k\pi, 2.80176 + 2k\pi$

47. $((2k + 1)\pi, -2)$ **49.** $\left(\frac{\pi}{3} + k\pi, \sqrt{3}\right)$





51. $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$

53. $\frac{\pi}{9}, \frac{2\pi}{9}, \frac{7\pi}{9}, \frac{8\pi}{9}, \frac{13\pi}{9}, \frac{14\pi}{9}$

55. $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$ **57.** 0 **59.** $\frac{k\pi}{2}$

61. $\frac{\pi}{9} + \frac{2k\pi}{3}, \frac{\pi}{2} + k\pi, \frac{5\pi}{9} + \frac{2k\pi}{3}$

63. $0, \pm 0.95$ **65.** 1.92 **67.** ± 0.71

Sección 8.6 ■

1. $\sqrt{2}\left(\cos\frac{\pi}{4} + i\operatorname{sen}\frac{\pi}{4}\right)$ **3.** $2\left(\cos\frac{7\pi}{4} + i\operatorname{sen}\frac{7\pi}{4}\right)$

5. $4\left(\cos\frac{11\pi}{6} + i\operatorname{sen}\frac{11\pi}{6}\right)$ **7.** $\sqrt{2}\left(\cos\frac{3\pi}{2} + i\operatorname{sen}\frac{3\pi}{2}\right)$

9. $5\sqrt{2}\left(\cos\frac{\pi}{4} + i\operatorname{sen}\frac{\pi}{4}\right)$

11. $8\left(\cos\frac{11\pi}{6} + i\operatorname{sen}\frac{11\pi}{6}\right)$ **13.** $20(\cos\pi + i\operatorname{sen}\pi)$

15. $5[\cos(\tan^{-1}\frac{4}{3}) + i\operatorname{sen}(\tan^{-1}\frac{4}{3})]$

17. $3\sqrt{2}\left(\cos\frac{3\pi}{4} + i\operatorname{sen}\frac{3\pi}{4}\right)$

19. $8\left(\cos\frac{\pi}{6} + i\operatorname{sen}\frac{\pi}{6}\right)$

21. $\sqrt{5}\,[\cos(\tan^{-1}\frac{1}{2}) + i\operatorname{sen}(\tan^{-1}\frac{1}{2})]$

23. $2\left(\cos\frac{\pi}{4} + i\operatorname{sen}\frac{\pi}{4}\right)$

25. $z_1z_2 = \cos\pi + i\operatorname{sen}\pi;\ \frac{z_1}{z_2} = \cos\left(\frac{\pi}{2}\right) - i\operatorname{sen}\left(\frac{\pi}{2}\right)$

27. $z_1z_2 = 14\left(\cos\frac{12\pi}{7} + i\operatorname{sen}\frac{12\pi}{7}\right)$

$\frac{z_1}{z_2} = \frac{7}{2}\left(\cos\frac{6\pi}{7} + i\operatorname{sen}\frac{6\pi}{7}\right)$

29. $z_1z_2 = 100(\cos 350° + i\operatorname{sen} 350°)$

$\frac{z_1}{z_2} = \frac{4}{25}(\cos 50° + i\operatorname{sen} 50°)$

31. $z_1 = 2\left(\cos\frac{\pi}{6} + i\operatorname{sen}\frac{\pi}{6}\right)$

$z_2 = 2\left(\cos\frac{\pi}{3} + i\operatorname{sen}\frac{\pi}{3}\right)$

$z_1z_2 = 4\left(\cos\frac{\pi}{2} + i\operatorname{sen}\frac{\pi}{2}\right)$

$\frac{z_1}{z_2} = \cos\frac{\pi}{6} - i\operatorname{sen}\frac{\pi}{6}$

$\frac{1}{z_1} = \frac{1}{2}\left(\cos\frac{\pi}{6} - i\operatorname{sen}\frac{\pi}{6}\right)$

33. $z_1 = 4\left(\cos\frac{11\pi}{6} + i\operatorname{sen}\frac{11\pi}{6}\right)$

$z_2 = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\operatorname{sen}\frac{3\pi}{4}\right)$

$z_1z_2 = 4\sqrt{2}\left(\cos\frac{7\pi}{12} + i\operatorname{sen}\frac{7\pi}{12}\right)$

$\frac{z_1}{z_2} = 2\sqrt{2}\left(\cos\frac{13\pi}{12} + i\operatorname{sen}\frac{13\pi}{12}\right)$

$\frac{1}{z_1} = \frac{1}{4}\left(\cos\frac{11\pi}{6} - i\operatorname{sen}\frac{11\pi}{6}\right)$

35. $z_1 = 5\sqrt{2}\left(\cos\frac{\pi}{4} + i\operatorname{sen}\frac{\pi}{4}\right)$

$z_2 = 4(\cos 0 + i\operatorname{sen} 0)$

$z_1z_2 = 20\sqrt{2}\left(\cos\frac{\pi}{4} + i\operatorname{sen}\frac{\pi}{4}\right)$

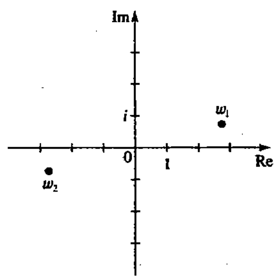
$$\frac{z_1}{z_2} = \frac{5\sqrt{2}}{4}\left(\cos\frac{\pi}{4} + i\,\text{sen}\,\frac{\pi}{4}\right)$$
$$\frac{1}{z_1} = \frac{\sqrt{2}}{10}\left(\cos\frac{\pi}{4} - i\,\text{sen}\,\frac{\pi}{4}\right)$$

37. $z_1 = 20(\cos\pi + i\,\text{sen}\,\pi)$
$$z_2 = 2\left(\cos\frac{\pi}{6} + i\,\text{sen}\,\frac{\pi}{6}\right)$$
$$z_1z_2 = 40\left(\cos\frac{7\pi}{6} + i\,\text{sen}\,\frac{7\pi}{6}\right)$$
$$\frac{z_1}{z_2} = 10\left(\cos\frac{5\pi}{6} + i\,\text{sen}\,\frac{5\pi}{6}\right)$$
$$\frac{1}{z_1} = \tfrac{1}{20}(\cos\pi - i\,\text{sen}\,\pi)$$

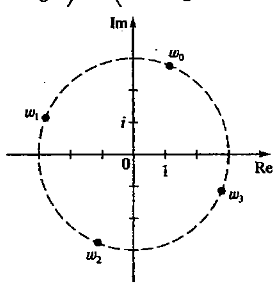
39. $-1{,}024$ **41.** $512(-\sqrt{3} + i)$ **43.** -1

45. $4{,}096$ **47.** $8(-1 + i)$ **49.** $\frac{1}{2{,}048}(-\sqrt{3} - i)$

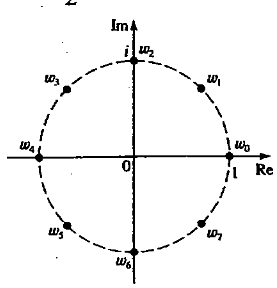
51. $2\sqrt{2}\left(\cos\frac{\pi}{12} + i\,\text{sen}\,\frac{\pi}{12}\right)$,
$2\sqrt{2}\left(\cos\frac{13\pi}{12} + i\,\text{sen}\,\frac{13\pi}{12}\right)$



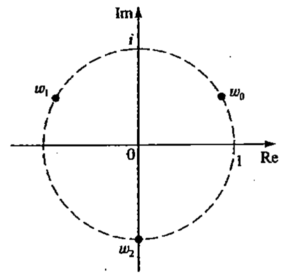
53. $3\left(\cos\frac{3\pi}{8} + i\,\text{sen}\,\frac{3\pi}{8}\right)$, $3\left(\cos\frac{7\pi}{8} + i\,\text{sen}\,\frac{7\pi}{8}\right)$,
$3\left(\cos\frac{11\pi}{8} + i\,\text{sen}\,\frac{11\pi}{8}\right)$, $3\left(\cos\frac{15\pi}{8} + i\,\text{sen}\,\frac{15\pi}{8}\right)$



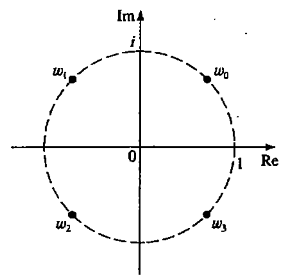
55. $\pm1,\ \pm i,\ \pm\frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}i$



57. $\frac{\sqrt{3}}{2} + \frac{1}{2}i,\ -\frac{\sqrt{3}}{2} + \frac{1}{2}i,\ -i$



59. $\pm\frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}i$



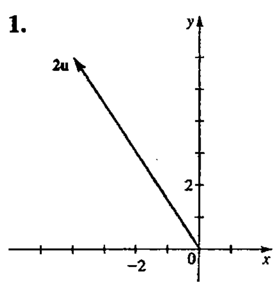
61. $\pm\frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}i$

63. $2\left(\cos\frac{\pi}{18} + i\,\text{sen}\,\frac{\pi}{18}\right)$, $2\left(\cos\frac{13\pi}{18} + i\,\text{sen}\,\frac{13\pi}{18}\right)$,
$2\left(\cos\frac{25\pi}{18} + i\,\text{sen}\,\frac{25\pi}{18}\right)$

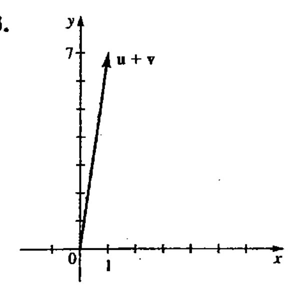
65. $2^{1/6}\left(\cos\frac{5\pi}{12} + i\,\text{sen}\,\frac{5\pi}{12}\right)$, $2^{1/6}\left(\cos\frac{13\pi}{12} + i\,\text{sen}\,\frac{13\pi}{12}\right)$,
$2^{1/6}\left(\cos\frac{21\pi}{12} + i\,\text{sen}\,\frac{21\pi}{12}\right)$

Sección 8.7 ■

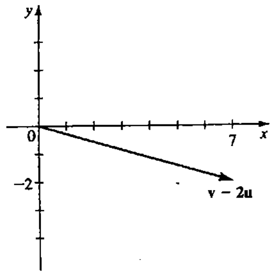
1.



3.



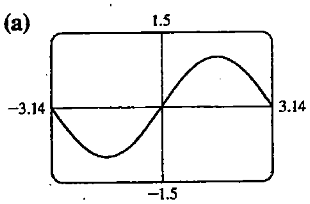
5.



7. $\langle 3,3\rangle$ **9.** $\langle 3,-1\rangle$ **11.** $\langle 5,7\rangle$ **13.** $\langle -4,-3\rangle$
15. $\langle 0,2\rangle$ **17.** $\langle 4,14\rangle, \langle -9,-3\rangle, \langle 5,8\rangle, \langle -6,17\rangle$
19. $\langle 0,-2\rangle, \langle 6,0\rangle, \langle -2,-1\rangle, \langle 8,-3\rangle$
21. $4\mathbf{i}, -9\mathbf{i}+6\mathbf{j}, 5\mathbf{i}-2\mathbf{j}, -6\mathbf{i}+8\mathbf{j}$
23. $\sqrt{5}, \sqrt{13}, 2\sqrt{5}, \frac{1}{2}\sqrt{13}, \sqrt{26}, \sqrt{10}, \sqrt{5}-\sqrt{13}$
25. $\sqrt{101}, 2\sqrt{2}, 2\sqrt{101}, \sqrt{2}, \sqrt{73}, \sqrt{145}, \sqrt{101}-2\sqrt{2}$
27. $20\sqrt{3}\,\mathbf{i}+20\mathbf{j}$ **29.** $-\frac{\sqrt{2}}{2}\mathbf{i}-\frac{\sqrt{2}}{2}\mathbf{j}$
31. $4\cos 10°\mathbf{i} + 4\,\text{sen}\,10°\mathbf{j} \approx 3.94\mathbf{i}+0.69\mathbf{j}$
33. $15\sqrt{3}, -15$ **35.** 5, 53.13° **37.** 13, 157.38°
39. 2, 60°
41. (a) $425\mathbf{i}+40\mathbf{j}$ (b) 427 mi/h, N84.6° E
43. 794 mi/h, N26.6° W
45. (a) $20\mathbf{i}+17.32\mathbf{j}$ (b) 26.5 mi/h, N49.1° E
47. 7.4 mi/h, 22.8 mi/h **49.** (a) $\langle 5,-3\rangle$ (b) $\langle -5,3\rangle$
51. (a) $-4\mathbf{j}$ (b) $4\mathbf{j}$
53. (a) $\langle -7.57, 10.61\rangle$ (b) $\langle 7.57, -10.61\rangle$
55. $\mathbf{T}_1 \approx -56.5\mathbf{i}+67.4\mathbf{j}, \mathbf{T}_2 \approx 56.5\mathbf{i}+32.6\mathbf{j}$

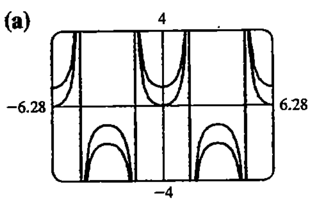
Repaso del capítulo 8 ■

23. (a)



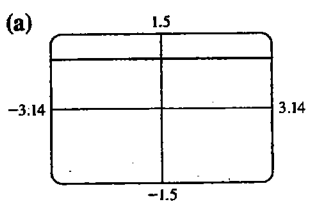
(b) Sí

25. (a)



(b) No

27. (a)

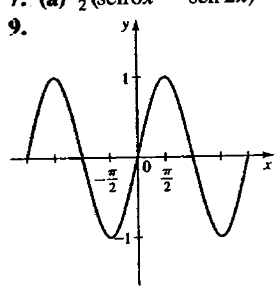


$2\,\text{sen}^2 3x + \cos 6x = 1$

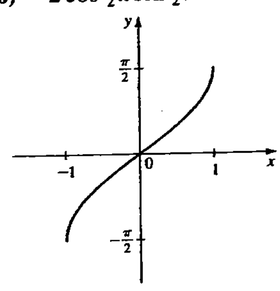
29. $0, \pi$ **31.** $\frac{\pi}{6}, \frac{5\pi}{6}$ **33.** $\frac{\pi}{3}, \frac{5\pi}{3}$ **35.** $\frac{2\pi}{3}, \frac{4\pi}{3}$
37. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{4}$
39. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$ **41.** $\frac{\pi}{6}$ **43.** 1.18
45. $\frac{1}{2}\sqrt{2+\sqrt{3}}$ **47.** $\sqrt{2}-1$ **49.** $\sqrt{2}/2$
51. $\sqrt{2}/2$ **53.** $\frac{\sqrt{2}+\sqrt{3}}{4}$ **55.** $2\frac{\sqrt{10}+1}{9}$
57. $\frac{2}{3}(\sqrt{2}+\sqrt{5})$ **59.** $\sqrt{(3+2\sqrt{2})/6}$ **61.** $\pi/3$
63. $\frac{1}{2}$ **65.** $2/\sqrt{21}$ **67.** $\frac{7}{9}$ **69.** $x/\sqrt{1+x^2}$
71. $\theta = \cos^{-1}\frac{x}{3}$ **73.** $s = 7{,}920\cos^{-1}\left(\frac{3{,}960}{h+3{,}960}\right)$
75. $4\sqrt{2}\left(\cos\frac{\pi}{4} + i\,\text{sen}\,\frac{\pi}{4}\right)$
77. $\sqrt{34}\left[\cos(\tan^{-1}\frac{3}{5}) + i\,\text{sen}(\tan^{-1}\frac{3}{5})\right]$
79. $\sqrt{2}\left(\cos\frac{3\pi}{4} + i\,\text{sen}\,\frac{3\pi}{4}\right)$ **81.** $8(-1+i\sqrt{3})$
83. $-\frac{1}{32}(1+i\sqrt{3})$ **85.** $\pm 2\sqrt{2}(1-i)$
87. $\pm 1, \pm\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$
89. $\sqrt{13}, \langle 6,4\rangle, \langle -10,2\rangle, \langle -4,6\rangle, \langle -22,7\rangle$
91. $3\mathbf{i}-4\mathbf{j}$ **93.** $(10,-2)$
95. (a) $(4.8\mathbf{i}+0.4\mathbf{j}) \times 10^4$
(b) 4.8×10^4 lb, N85.2° E

Examen del capítulo 8 ■

2. (a) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (b) $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$
3. 0.57964, 2.56195, 3.72123, 5.70355 **4.** $\tan\theta$
5. (a) $\frac{1}{2}$ (b) $\frac{\sqrt{2}+\sqrt{6}}{4}$ (c) $\frac{1}{2}\sqrt{2-\sqrt{3}}$
6. $(10-2\sqrt{5})/15$
7. (a) $\frac{1}{2}(\text{sen}\,8x - \text{sen}\,2x)$ (b) $-2\cos\frac{7}{2}x\,\text{sen}\,\frac{3}{2}x$ **8.** -2
9.



Dominio ℝ

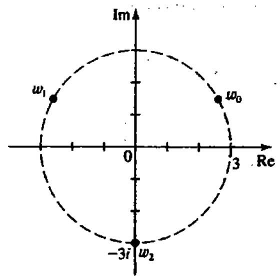


Dominio $[-1, 1]$

10. (a) $\theta = \tan^{-1}\frac{x}{4}$ (b) $\theta = \cos^{-1}\frac{3}{x}$ **11.** $\frac{40}{41}$
12. (a) $2\left(\cos\frac{\pi}{3} + i\,\text{sen}\,\frac{\pi}{3}\right)$ (b) -512

13. $-8, \sqrt{3}+i$

14. $-3i, 3\left(\pm\dfrac{\sqrt{3}}{2}+\dfrac{1}{2}i\right)$



15. **(a)** $-6\mathbf{i}+10\mathbf{j}$ **(b)** $2\sqrt{34}$

16. **(a)** $\langle 19, -3\rangle$ **(b)** $5\sqrt{2}$

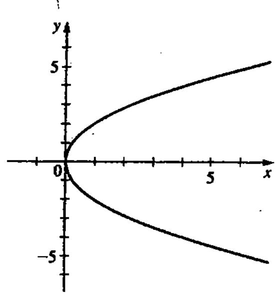
17. **(a)** $14\mathbf{i}+6\sqrt{3}\mathbf{j}$ **(b)** 17.4 mi/h, N53.4°E

CAPÍTULO 9

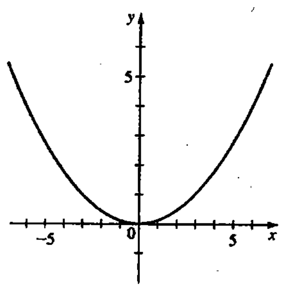
Sección 9.1 ■

Orden de las respuestas: foco, directriz, diámetro focal

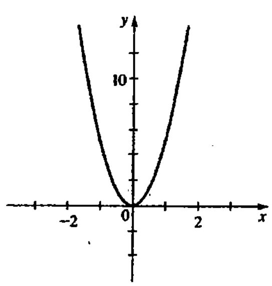
1. $F(1,0);\ x=-1;\ 4$



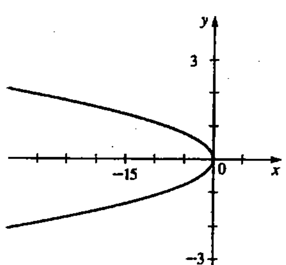
3. $F(0,\frac{9}{4});\ y=-\frac{9}{4};\ 9$



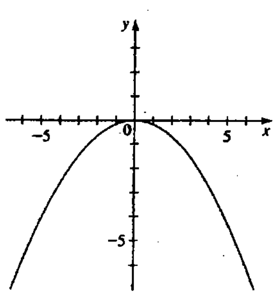
5. $F(0,\frac{1}{20});\ y=-\frac{1}{20};\ \frac{1}{5}$



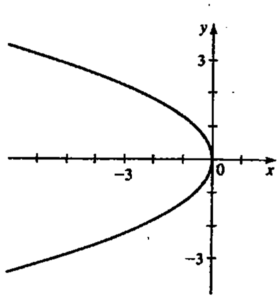
7. $F(-\frac{1}{32},0);\ x=\frac{1}{32};\ \frac{1}{8}$



9. $F(0,-\frac{3}{2});\ y=\frac{3}{2};\ 6$



11. $F(-\frac{5}{12},0);\ x=\frac{5}{12};\ \frac{5}{3}$



13. $x^2=8y$ **15.** $y^2=-32x$

17. $y^2=-8x$ **19.** $x^2=40y$

21. $y^2=4x$ **23.** $x^2=20y$

25. $x^2=-12y$ **27.** $y^2=-3x$

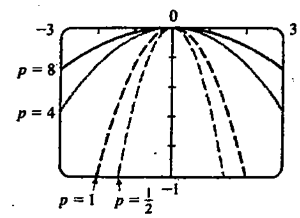
29. $x=y^2$ **31.** $x^2=-4\sqrt{2}\,y$

33. **(a)** $y^2=12x$ **(b)** $8\sqrt{15}\approx 31$ cm

35. $x^2=600y$

37. **(a)** $x^2=-4py,\ p=\frac{1}{2}$, 1, 4, y 8

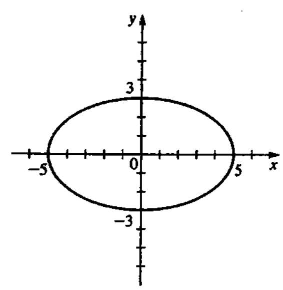
(b) Mientras más cercana está la directriz al vértice, la parábola es más cerrada.



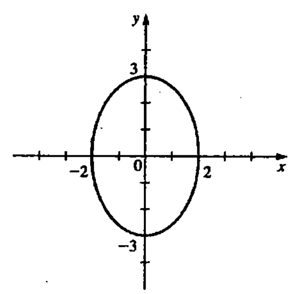
Sección 9.2 ■

Orden de las respuestas: vértices, focos, excentricidad, eje mayor y eje menor

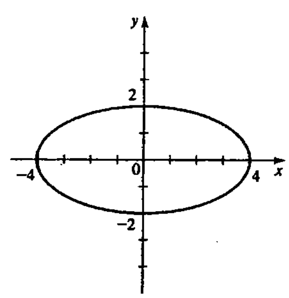
1. $V(\pm5,0);\ F(\pm4,0);$ $\frac{4}{5};\ 10,\ 6$



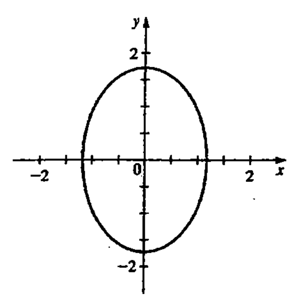
3. $V(0,\pm3);\ F(0,\pm\sqrt{5});$ $\sqrt{5}/3;\ 6,\ 4$



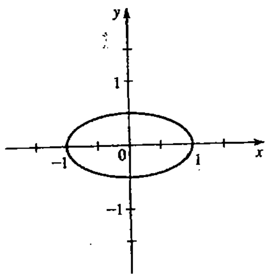
5. $V(\pm4,0);\ F(\pm2\sqrt{3},0);$ $\sqrt{3}/2;\ 8,\ 4$



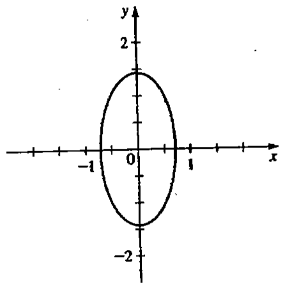
7. $V(0,\pm\sqrt{3});\ F(0,\pm\sqrt{3/2});$ $1/\sqrt{2};\ 2\sqrt{3},\ \sqrt{6}$



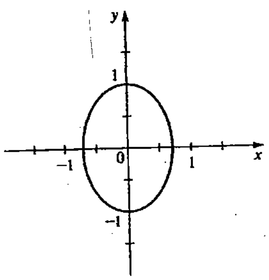
9. $V(\pm 1, 0)$; $F(\pm\sqrt{3}/2, 0)$; $\sqrt{3}/2$; 2, 1



11. $V(0, \pm\sqrt{2})$; $F(0, \pm\sqrt{3/2})$; $\sqrt{3}/2$; $2\sqrt{2}$, $\sqrt{2}$



13. $V(0, \pm 1)$; $F(0, \pm 1/\sqrt{2})$; $1/\sqrt{2}$; 2, $\sqrt{2}$



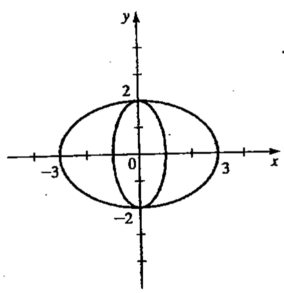
15. $\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$ **17.** $\dfrac{x^2}{4} + \dfrac{y^2}{8} = 1$

19. $\dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$ **21.** $x^2 + \dfrac{y^2}{4} = 1$

23. $\dfrac{x^2}{9} + \dfrac{y^2}{13} = 1$ **25.** $\dfrac{x^2}{100} + \dfrac{y^2}{91} = 1$

27. $\dfrac{x^2}{25} + \dfrac{y^2}{5} = 1$ **29.** $\dfrac{64x^2}{225} + \dfrac{64y^2}{81} = 1$

31. $(0, \pm 2)$

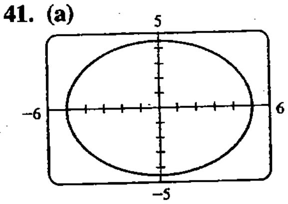


33. $\dfrac{x^2}{2.2500 \times 10^{16}} + \dfrac{y^2}{2.2497 \times 10^{16}} = 1$

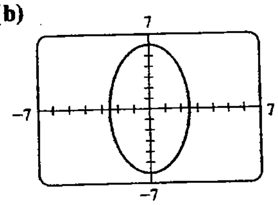
35. $\dfrac{x^2}{1{,}455{,}642} + \dfrac{y^2}{1{,}451{,}610} = 1$

37. $5\sqrt{39}/2 \approx 15.6$ pulg

41. (a)



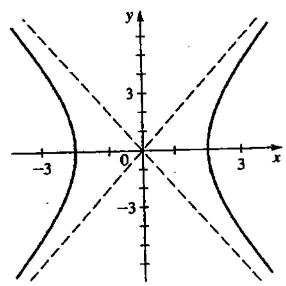
(b)



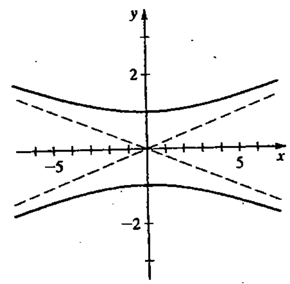
Sección 9.3

Orden de las respuestas: vértices, focos, asíntotas

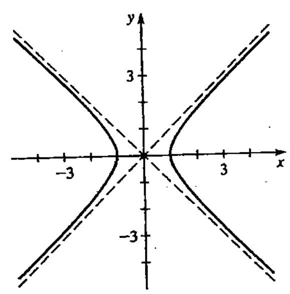
1. $V(\pm 2, 0)$; $F(\pm 2\sqrt{5}, 0)$; $y = \pm 2x$



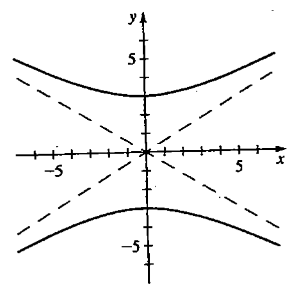
3. $V(0, \pm 1)$; $F(0, \pm\sqrt{26})$; $y = \pm\frac{1}{5}x$



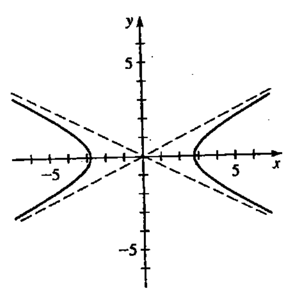
5. $V(\pm 1, 0)$; $F(\pm\sqrt{2}, 0)$; $y = \pm x$



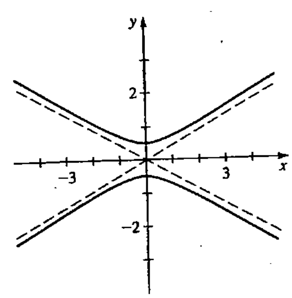
7. $V(0, \pm 3)$; $F(0, \pm\sqrt{34})$; $y = \pm\frac{3}{5}x$



9. $V(\pm 2\sqrt{2}, 0)$; $F(\pm\sqrt{10}, 0)$; $y = \pm\frac{1}{2}x$



11. $V(0, \pm\frac{1}{2})$; $F(0, \pm\sqrt{5}/2)$; $y = \pm\frac{1}{2}x$



13. $\dfrac{x^2}{4} - \dfrac{y^2}{12} = 1$ **15.** $\dfrac{y^2}{16} - \dfrac{x^2}{16} = 1$

17. $\dfrac{x^2}{9} - \dfrac{y^2}{16} = 1$ **19.** $y^2 - \dfrac{x^2}{3} = 1$

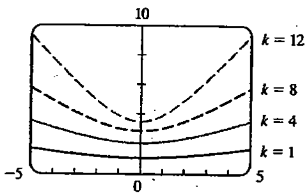
21. $x^2 - \dfrac{y^2}{25} = 1$ **23.** $\dfrac{5y^2}{64} - \dfrac{5x^2}{256} = 1$

25. $\dfrac{x^2}{16} - \dfrac{y^2}{16} = 1$ **27.** $\dfrac{x^2}{9} - \dfrac{y^2}{16} = 1$

29. **(b)** $x^2 - y^2 = c^2/2$

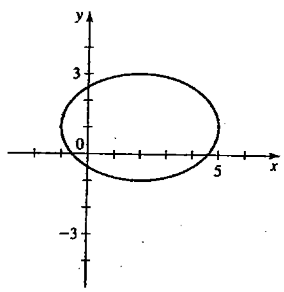
33. **(a)** 490 mi **(b)** $\dfrac{y^2}{60{,}025} - \dfrac{x^2}{2{,}475} = 1$ **(c)** 10.1 mi

35. **(b)**

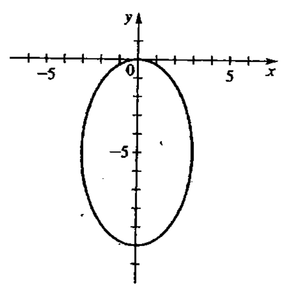


Sección 9.4 ■

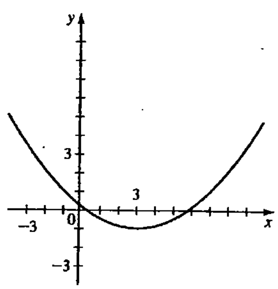
1. Centro $C(2, 1)$; focos $F(2 \quad \sqrt{5}, 1)$; vértices $V_1(-1, 1)$, $V_2(5, 1)$; eje mayor 6, eje menor 4



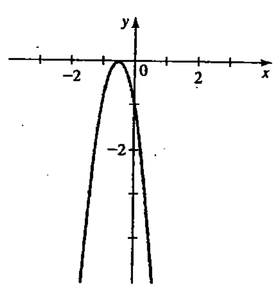
3. Centro $C(0, -5)$; focos $F_1(0, -1)$, $F_2(0, -9)$; vértices $V_1(0, 0)$, $V_2(0, -10)$; eje mayor 10, eje menor 6



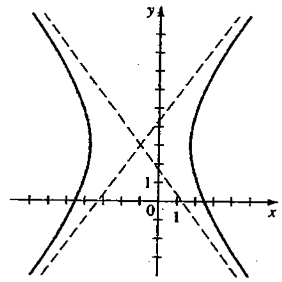
5. Vértice $V(3, -1)$; foco $F(3, 1)$; directriz $y = -3$



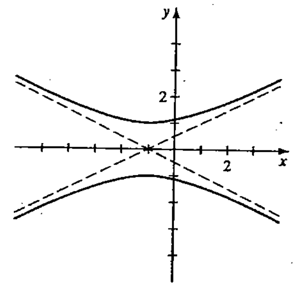
7. Vértice $V\left(-\frac{1}{2}, 0\right)$; foco $F\left(-\frac{1}{2}, -\frac{1}{16}\right)$; directriz $y = \frac{1}{16}$



9. Centro $C(-1, 3)$; focos $F_1(-6, 3)$, $F_2(4, 3)$; vértices $V_1(-4, 3)$, $V_2(2, 3)$; asíntotas
$$y = \pm\tfrac{4}{3}(x + 1) + 3$$



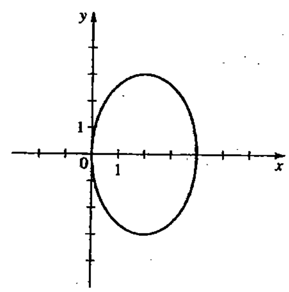
11. Centro $C(-1, 0)$; focos $F(-1, \quad \sqrt{5})$; vértices $V(-1, 1)$; asíntotas
$$y = \pm\tfrac{1}{2}(x + 1)$$



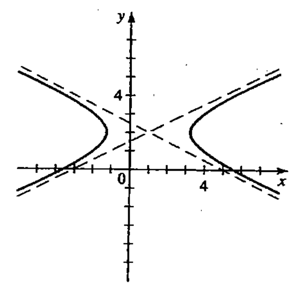
13. $x^2 = -\frac{1}{4}(y - 4)$ **15.** $\dfrac{(x - 5)^2}{25} + \dfrac{y^2}{16} = 1$

17. $(y - 1)^2 - x^2 = 1$

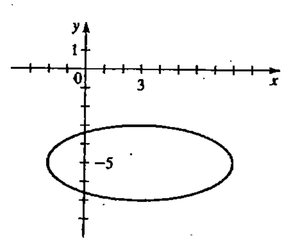
19. Elipse; $C(2, 0)$; $F(2, \pm\sqrt{5})$; $V(2, \pm 3)$; eje mayor 6, eje menor 4



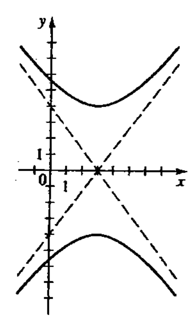
21. Hipérbola; $C(1, 2)$; $F_1\left(-\frac{3}{2}, 2\right)$, $F_2\left(\frac{7}{2}, 2\right)$; $V\left(1 \pm \sqrt{5}, 2\right)$; asíntotas
$$y = \pm\tfrac{1}{2}(x - 1) + 2$$



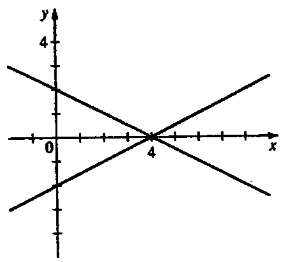
23. Elipse; $C(3, -5)$; $F(3 \pm \sqrt{21}, -5)$; $V_1(-2, -5)$, $V_1(8, -5)$; eje mayor 10, eje menor 4



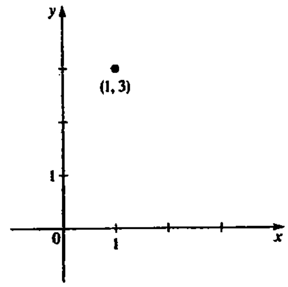
25. Hipérbola; $C(3, 0)$; $F(3, \pm 5)$; $V(3, \pm 4)$; asíntotas
$$y = \pm\tfrac{4}{3}(x - 3)$$



27. Cónica degenerada (par de rectas), $y = \pm\frac{1}{2}(x - 4)$

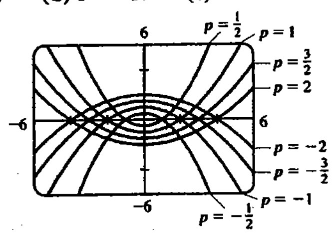


29. Punto $(1, 3)$



31. **(a)** $F < 17$ **(b)** $F = 17$ **(c)** $F > 17$

33. **(a)**



(c) Las parábolas se angostan.

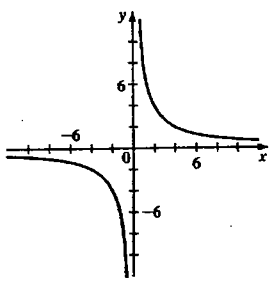
Sección 9.5 ■

1. $(\sqrt{2}, 0)$ **3.** $(0, -2\sqrt{3})$ **5.** $(1.6383, 1.1472)$

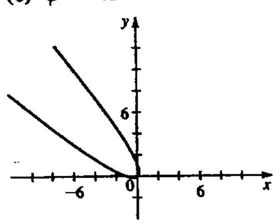
7. $X^2 + Y^2 - 2XY - 3\sqrt{2}\,X + \sqrt{2}\,Y + 2 = 0$

9. $X^2 - Y^2 = 2$

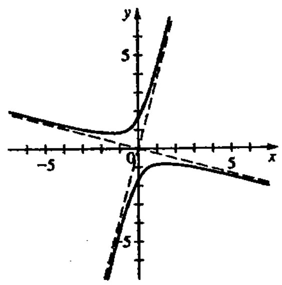
11. **(a)** Hipérbola **(b)** $X^2 - Y^2 = 16$
(c) $\phi = 45°$



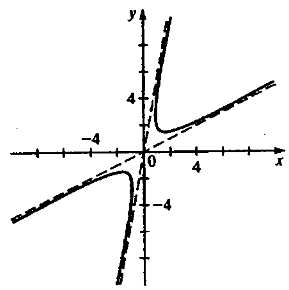
13. **(a)** Parábola **(b)** $Y = \sqrt{2}\,X^2$
(c) $\phi = 45°$



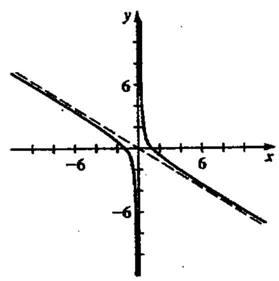
15. **(a)** Hipérbola
(b) $Y^2 - X^2 = 1$
(c) $\phi = 30°$



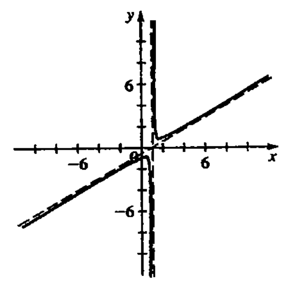
17. **(a)** Hipérbola
(b) $\dfrac{X^2}{4} - Y^2 = 1$
(c) $\phi \approx 53°$



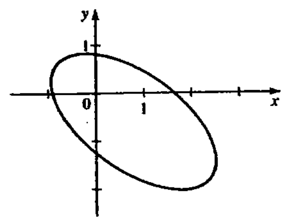
19. **(a)** Hipérbola
(b) $3X^2 - Y^2 = 2\sqrt{3}$
(c) $\phi = 30°$



21. **(a)** Hipérbola
(b) $(X - 1)^2 - 3Y^2 = 1$
(c) $\phi = 60°$



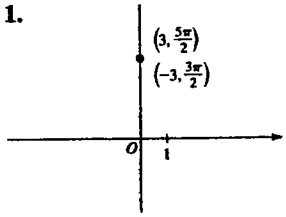
23. **(a)** Elipse
(b) $X^2 + \dfrac{(Y + 1)^2}{4} = 1$
(c) $\phi \approx 53°$
Véase la gráfica a la derecha



25. **(a)** $(X - 5)^2 - Y^2 = 1$
(b) Coordenadas XY: $C(5, 0)$; $V_1(6, 0)$, $V_2(4, 0)$; $F(5 \pm \sqrt{2}, 0)$; coordenadas xy: $C(4, 3)$; $V_1(\frac{24}{5}, \frac{18}{5})$, $V_2(\frac{16}{5}, \frac{12}{5})$; $F(4 \pm \frac{4}{5}\sqrt{2}, 3 \pm \frac{3}{5}\sqrt{2})$
(c) $Y = \pm(X - 5)$; $7x - y - 25 = 0$, $x + 7y - 25 = 0$

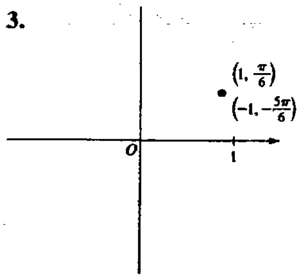
Sección 9.6 ■

1.

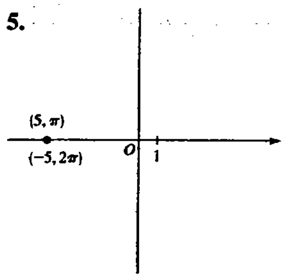


$\left(-3, \dfrac{3\pi}{2}\right), \left(3, \dfrac{5\pi}{2}\right)$

3.



5.



$\left(-1, -\dfrac{5\pi}{6}\right), \left(1, \dfrac{\pi}{6}\right)$ $\qquad (-5, 2\pi), (5, \pi)$

7. $(2\sqrt{3}, 2)$ **9.** $(1, -1)$ **11.** $(-5, 0)$

13. $\left(\sqrt{2}, \dfrac{3\pi}{4}\right)$ **15.** $\left(4, \dfrac{\pi}{4}\right)$

17. $(5, \tan^{-1}\frac{4}{3})$ **19.** $\theta = \dfrac{\pi}{4}$

21. $r = \tan\theta \sec\theta$ **23.** $r = 4\sec\theta$

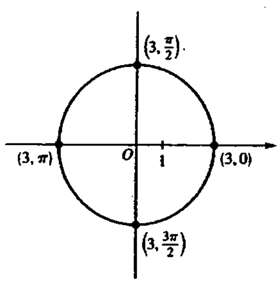
25. $x^2 + y^2 = 49$ **27.** $x = 6$

29. $x^2 + y^2 = \dfrac{y}{x}$ **31.** $y - x = 1$

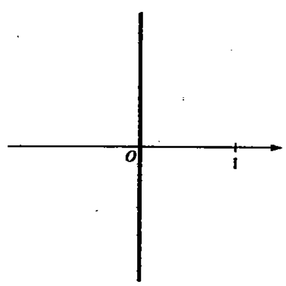
33. $x^2 + y^2 = (x^2 + y^2 - x)^2$

35. $x = 2$ **37.** $y = \pm\sqrt{3}\,x$

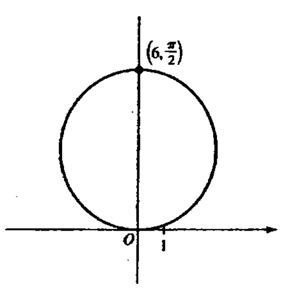
39.



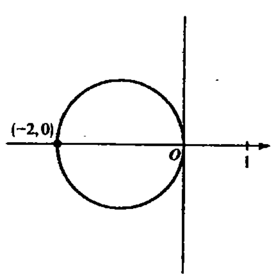
41.



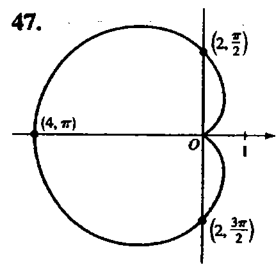
43.



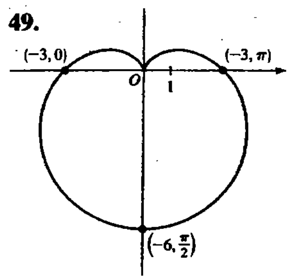
45.



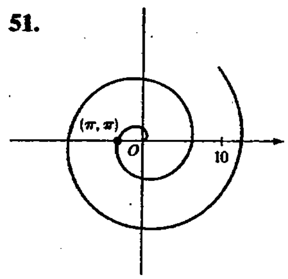
47.



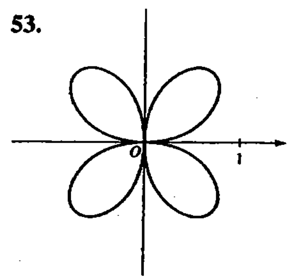
49.



51.



53.



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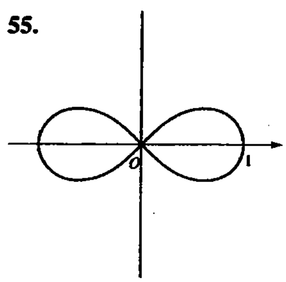
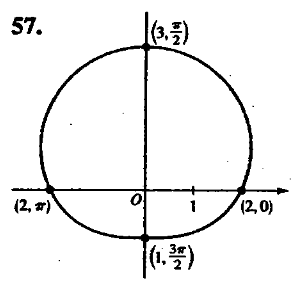
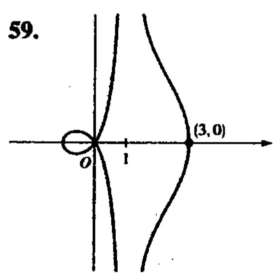


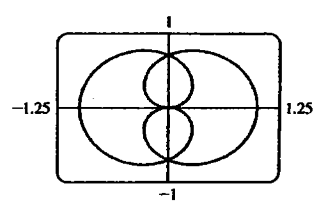
57.



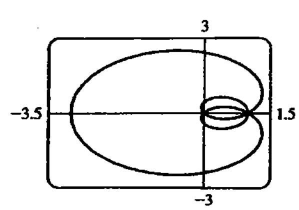
59.



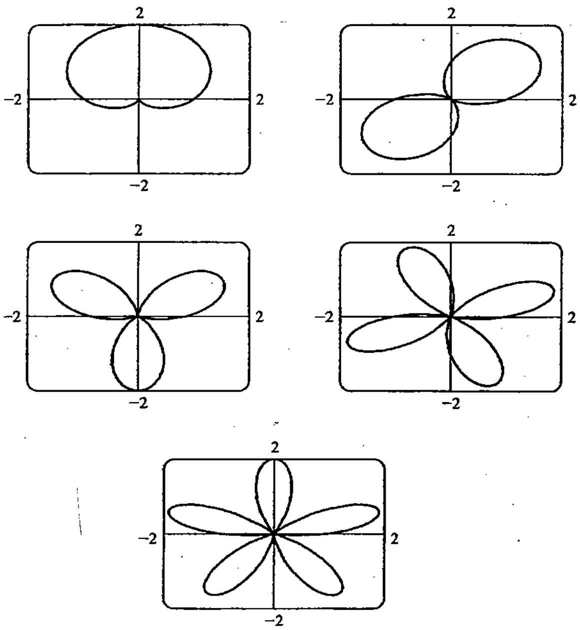
61. $0 \leq \theta \leq 4\pi$



63. $0 \leq \theta \leq 4\pi$



65.



La gráfica de $r = 1 + \operatorname{sen} n\theta$ tiene n bucles.

67. IV **69.** III **71.** **(b)** $\sqrt{10 + 6\cos(5\pi/12)} \approx 3.40$

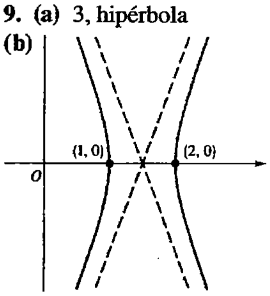
Sección 9.7 ■

1. $r = 6/(3 + 2\cos\theta)$ **3.** $r = 2/(1 + \operatorname{sen}\theta)$

5. $r = 20/(1 + 4\cos\theta)$ **7.** $r = 10/(1 + \operatorname{sen}\theta)$

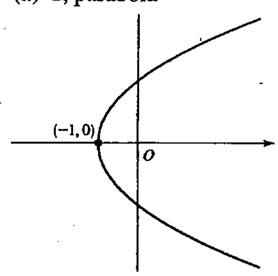
9. **(a)** 3, hipérbola

(b)



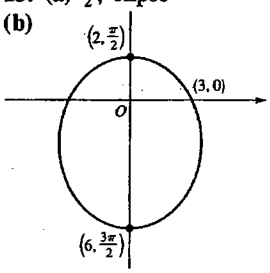
11. **(a)** 1, parábola

(b)



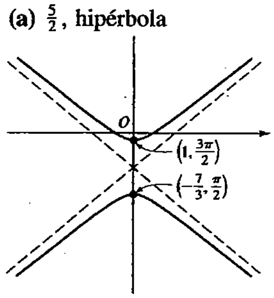
13. **(a)** $\frac{1}{2}$, elipse

(b)



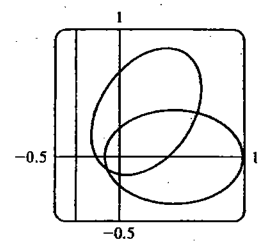
15. **(a)** $\frac{5}{2}$, hipérbola

(b)

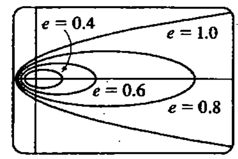


17. **(a)** $e = \frac{3}{4}$, directriz $x = -\frac{1}{3}$

(b) $r = \dfrac{1}{4 - 3\cos\left(\theta - \dfrac{\pi}{3}\right)}$



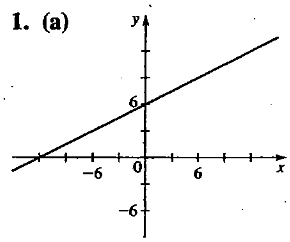
19. La elipse es casi circular cuando e es cercana a 0, y se vuelve más alargada a medida que $e \to 1^-$. En $e = 1$, la curva se transforma en parábola.



21. **(b)** $r = (1.49 \times 10^8)/(1 - 0.017\cos\theta)$ **23.** 0.25

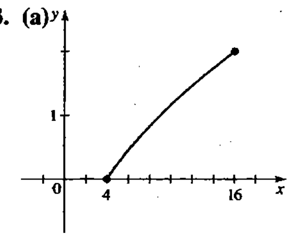
Sección 9.8 ■

1. **(a)**



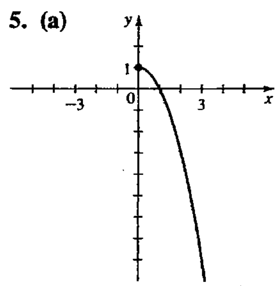
(b) $x - 2y + 12 = 0$

3. **(a)**



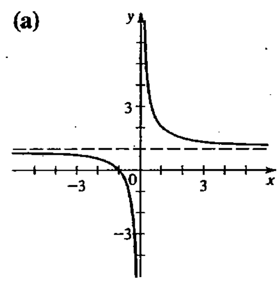
(b) $x = (y + 2)^2$

5. **(a)**



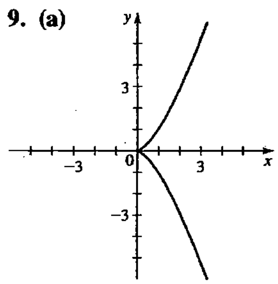
(b) $x = \sqrt{1 - y}$

7. **(a)**



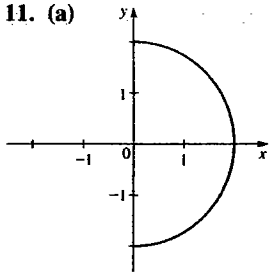
(b) $y = \dfrac{1}{x} + 1$

9. **(a)**



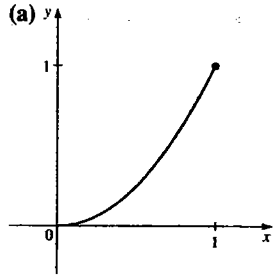
(b) $x^3 = y^2$

11. (a)



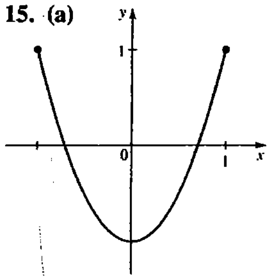
(b) $x^2 + y^2 = 4$

13. (a)



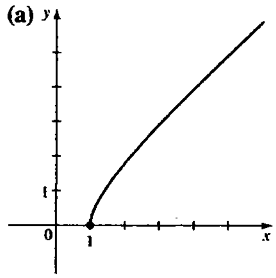
(b) $y = x^2$

15. (a)



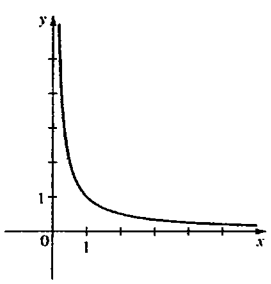
(b) $y = 2x^2 - 1$

17. (a)



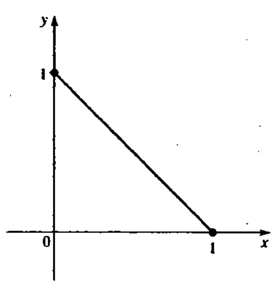
(b) $x^2 - y^2 = 1$

19. (a)



(b) $xy = 1$

21. (a)



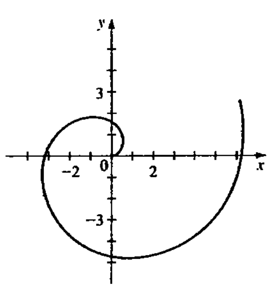
(b) $x + y = 1$

23. $x = 4 + t,\ y = -1 + \frac{1}{2}t$

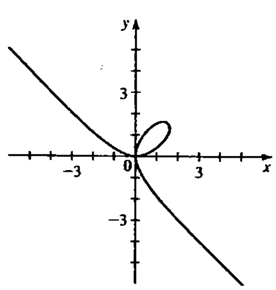
25. $x = 6 + t,\ y = 7 + t$

27. $x = a\cos t,\ y = a\operatorname{sen} t$

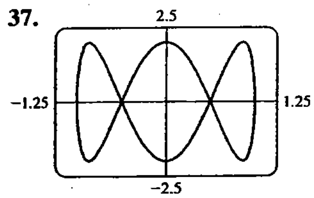
31.



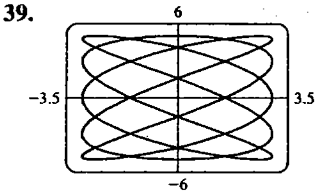
33.



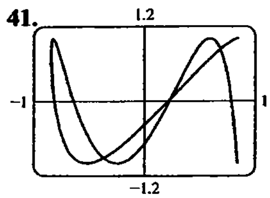
37.



39.

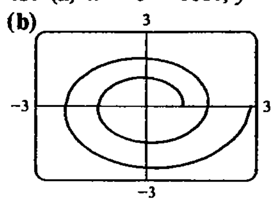


41.



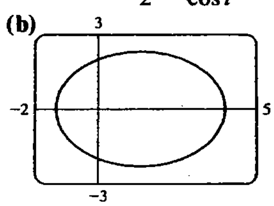
43. (a) $x = e^{t/12}\cos t,\ y = e^{t/12}\operatorname{sen} t$

(b)



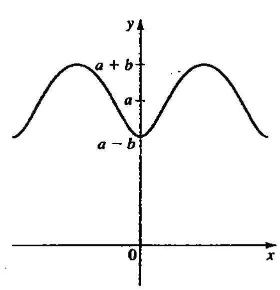
45. (a) $x = \dfrac{4\cos t}{2 - \cos t},\ y = \dfrac{4\operatorname{sen} t}{2 - \cos t}$

(b)

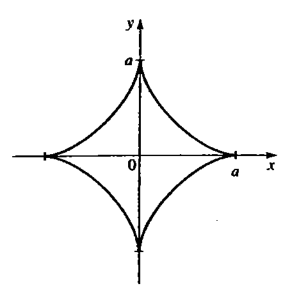


47. III **49.** II

51.



53. (b) $x^{2/3} + y^{2/3} = a^{2/3}$

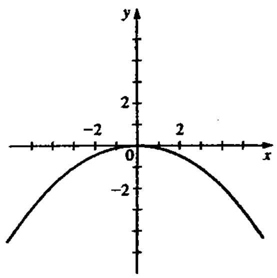


55. $x = a(\operatorname{sen}\theta\cos\theta + \cot\theta),\ y = a(1 + \operatorname{sen}^2\theta)$

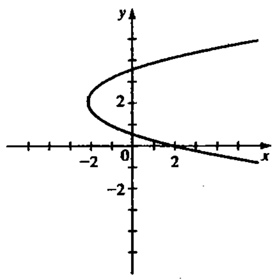
57. $y = a - a\cos\left(\dfrac{x + \sqrt{2ay - y^2}}{a}\right)$

Repaso del capítulo 9 ■

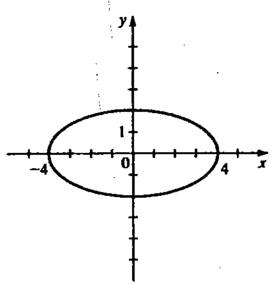
1. $V(0,0)$; $F(0,-2)$; $y = 2$



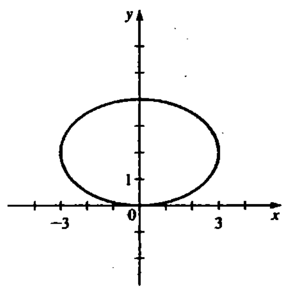
3. $V(-2,2)$; $F(-\frac{7}{4},2)$; $x = -\frac{9}{4}$



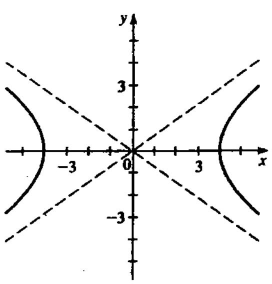
5. $C(0,0)$; $V(\pm 4,0)$; $F(\pm 2\sqrt{3},0)$; ejes 8, 4



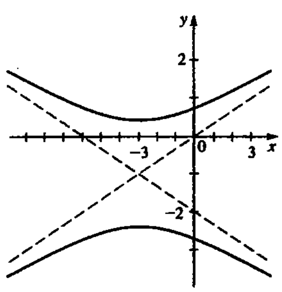
7. $C(0,2)$; $V(\pm 3,2)$; $F(\pm\sqrt{5},2)$; ejes 6, 4



9. $C(0,0)$; $V(\pm 4,0)$; $F(\pm 2\sqrt{6},0)$; asíntotas $y = \pm\dfrac{1}{\sqrt{2}}x$



11. $C(-3,-1)$; $V(-3,-1\pm\sqrt{2})$; $F(-3,-1\pm 2\sqrt{5})$; asíntotas $y = \frac{1}{3}x$, $y = -\frac{1}{3}x - 2$

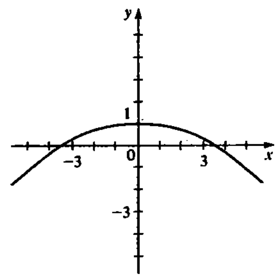


13. $y^2 = 8x$

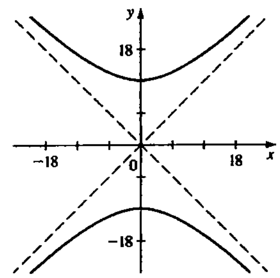
15. $\dfrac{y^2}{16} - \dfrac{x^2}{9} = 1$

17. $\dfrac{(x-4)^2}{16} + \dfrac{(y-2)^2}{4} = 1$

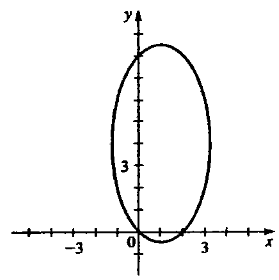
19. Parábola; $F(0,-2)$; $V(0,1)$



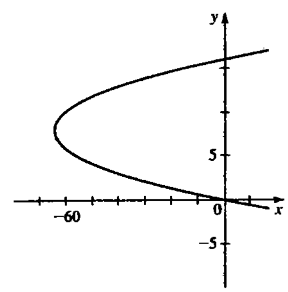
21. Hipérbola; $F(0,\pm 12\sqrt{2})$; $V(0,\pm 12)$



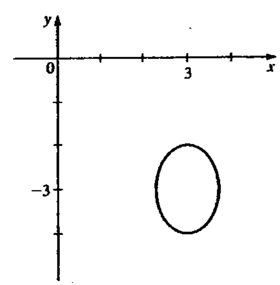
23. Elipse; $F(1,4\pm\sqrt{15})$; $V(1,4\pm 2\sqrt{5})$



25. Parábola; $F(-\frac{255}{4},8)$; $V(-64,8)$



27. Elipse; $F(3,-3\pm 1/\sqrt{2})$; $V_1(3,-4)$, $V_2(3,-2)$



29. No tiene gráfica **31.** $x^2 = 4y$ **33.** $\dfrac{y^2}{4} - \dfrac{x^2}{16} = 1$

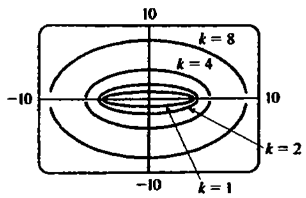
35. $\dfrac{(x-1)^2}{3} + \dfrac{(y-2)^2}{4} = 1$

37. $\dfrac{4(x-7)^2}{225} + \dfrac{(y-2)^2}{100} = 1$

39. $(x - 800)^2 = -200(y - 3{,}200)$

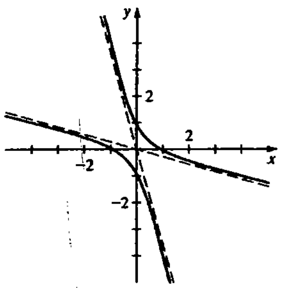
41. **(a)** 91,419,000 mi **(b)** 94,581,000 mi

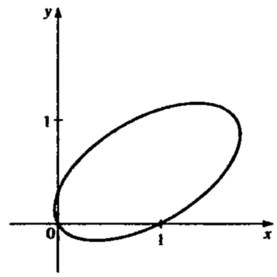
43. (a)



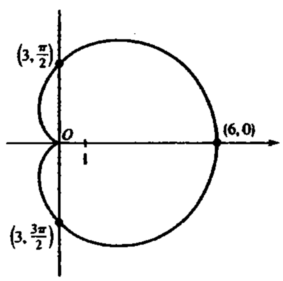
45. (a) Hipérbola
(b) $3X^2 - Y^2 = 1$
(c) $\phi = 45°$

47. (a) Elipse
(b) $(X - 1)^2 + 4Y^2 = 1$
(c) $\phi = 30°$



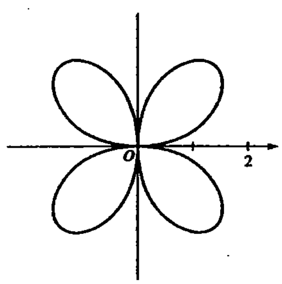


49. (a)



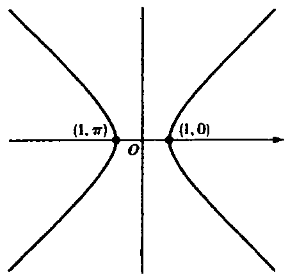
(b) $(x^2 + y^2 - 3x)^2 = 9(x^2 + y^2)$

51. (a)

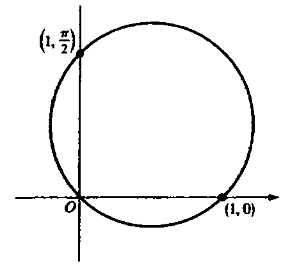


(b) $(x^2 + y^2)^3 = 16x^2y^2$

53. (a)



55. (a)

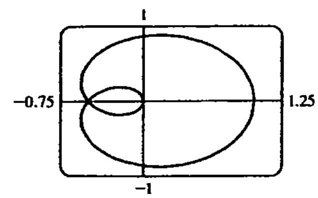


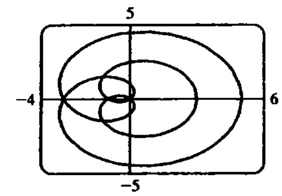
(b) $x^2 - y^2 = 1$

(b) $x^2 + y^2 = x + y$

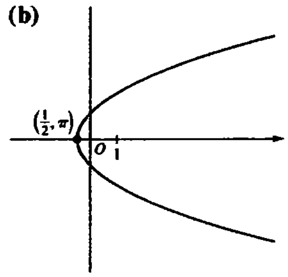
57. $0 \leqslant \theta \leqslant 6\pi$

59. $0 \leqslant \theta \leqslant 6\pi$

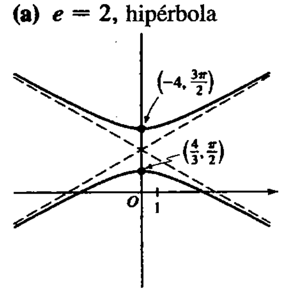




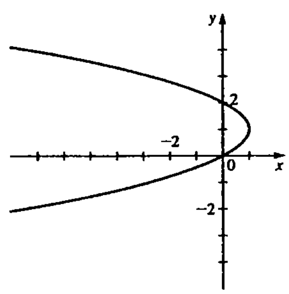
61. (a) $e = 1$, parábola
(b)



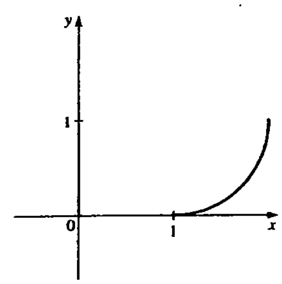
63. (a) $e = 2$, hipérbola
(b)



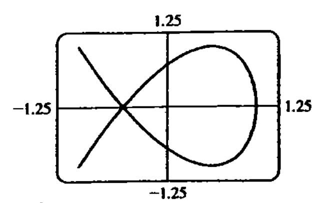
65. $x = 2y - y^2$



67. $(x - 1)^2 + (y - 1)^2 = 1$

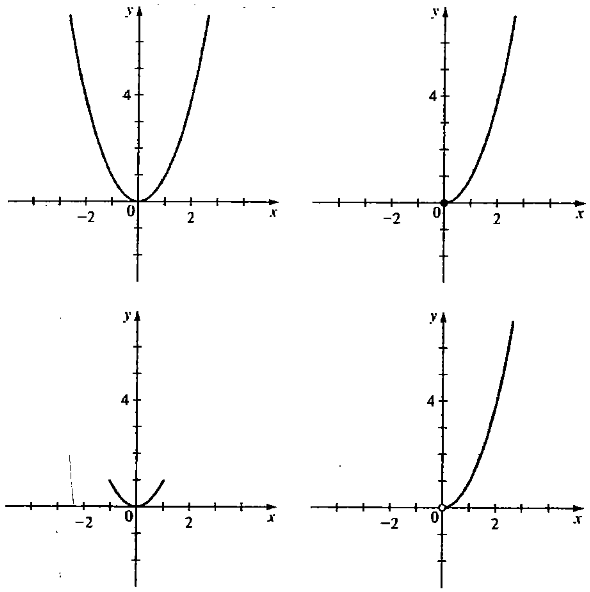


69.



71. **(a)** $y = x^2$

(b)

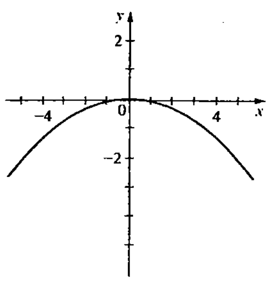


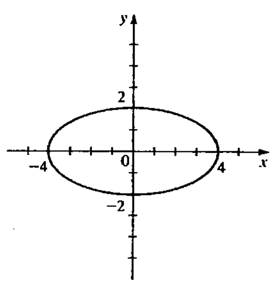
Las curvas son distintas partes de la parábola $y = x^2$.

Examen del capítulo 9 ■

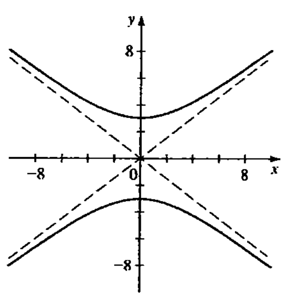
1. $F(0, -3)$, $y = 3$

2. $V(\pm 4, 0)$; $F(\pm 2\sqrt{3}, 0)$; 8, 4





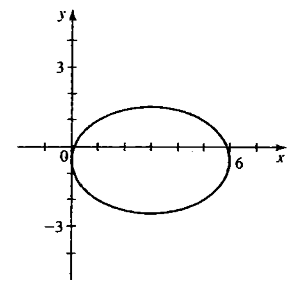
3. $V(0, \pm 3)$; $F(0, \pm 5)$; $y = \pm \frac{3}{4}x$



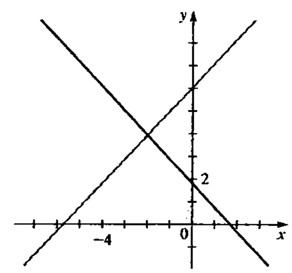
4. $y^2 = -x$ **5.** $\dfrac{x^2}{16} + \dfrac{(y-3)^2}{9} = 1$

6. $(x-2)^2 - \dfrac{y^2}{3} = 1$

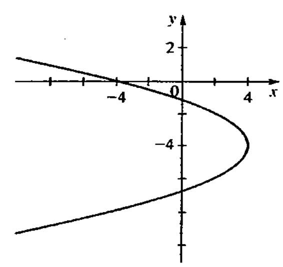
7. $\dfrac{(x-3)^2}{9} + \dfrac{\left(y+\frac{1}{2}\right)^2}{4} = 1$



8. $9(x+2)^2 - 8(y-4)^2 = 0$



9. $(y+4)^2 = -2(x-4)$

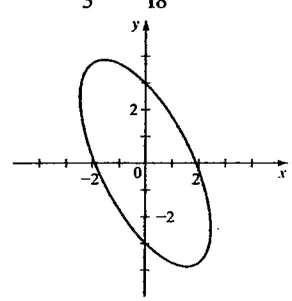


10. $\dfrac{y^2}{9} - \dfrac{x^2}{16} = 1$ **11.** $x^2 - 4x - 8y + 20 = 0$

12. $\frac{3}{4}$ pulg

13. **(a)** Elipse **(b)** $\dfrac{X^2}{3} + \dfrac{Y^2}{18} = 1$

(c) $\phi \approx 27°$



(d) $\left(-3\sqrt{2/5}, 6\sqrt{2/5}\right)$, $\left(3\sqrt{2/5}, -6\sqrt{2/5}\right)$